Probability & Statistics I

Week 5 — Algorithmic Approaches to Mathematics

Dhruva Raman

England, 1645

We all hate that we can't control uncertainty



Bad thing happens

Witch caused it?

Sink (innocent) / float (guilty)

Kill witch, problem solved

Causes are relieving

- Killing the witch will end my misfortune

Assumption:
Buoyancy as measure of guilt.
Verification?

Causes are relieving

- Killing the witch will end my misfortune

Assumption:
Buoyancy as measure of guilt.
Verification?

Quantitative 'causes' in our society:

- Doing a degree increases your lifetime earnings by £10000
- Taking this drug will decrease your cholesterol by 20%
- Decreasing your cholesterol decreases your heart attack risk by 30%

Assumption:
Hidden maths on some(?) cohort as measure of you.
Verification?

You are constantly being lied to with numbers

This should make you angry

What does statistics / data science actually mean

Method for answering why under uncertainty. That can be wrong!.

What does statistics / data science actually mean

Method for answering why under uncertainty. That can be wrong!.

- Taking this drug will decrease my cholesterol by 20%? Should I act?

Understand the data science, assumptions, and decide for yourself

Goals today (first half)

Probability

What's an experiment?

What's a probability?

What's a random variable?[1]

[1] It's neither random nor a variable

What is an experiment

A process with uncertain outcomes

Data scientist

- Measure population's credit scores

Climate scientist

- Measure tomorrow's weather

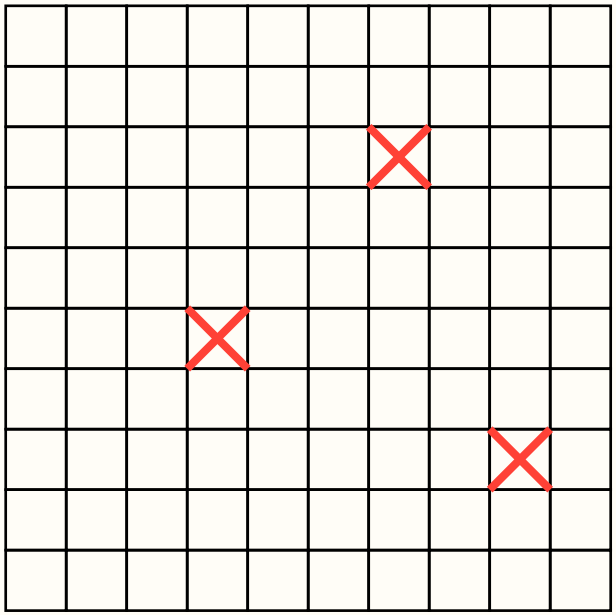
Goal of probability

Framework to mathematically characterise uncertain events

Purpose: quantify and compute with uncertainty

Uncertain lecture theatre seating next week

Possible



= empty seat

Terminology

Each *possible* seating combination is an outcome ω

The *outcome space* Ω is a set containing all possible outcomes[1]

$$\underbrace{\omega}_{\text{single outcome}} \quad \in \quad \underbrace{\Omega}_{\text{all outcomes}}$$

[1] Remember: set means *unordered* collection

Questions

How many outcomes in a coin flip?

Questions

How many outcomes in a coin flip?

- $\Omega = \{H, T\}$
- $| \Omega | = 2$

"{" brackets denote *sets*

$| \bullet |$ means *cardinality*: how many elements the thing inside has

What's the second outcome in a coin flip?

Questions

How many outcomes in a coin flip?

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What's the second outcome in a coin flip?

Invalid question

"{" brackets denote *sets*

$| \bullet |$ means *cardinality*: how many elements the thing inside has

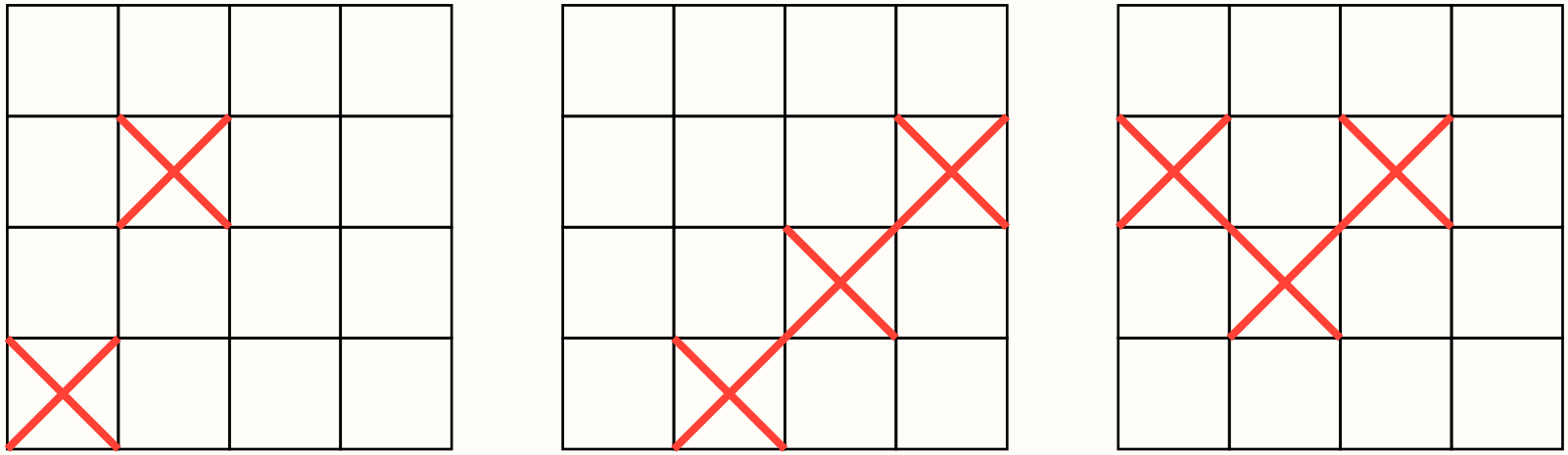
Sets are unordered

$\Omega = \{T, H\}$ as well

Events are sets of outcomes

Event E: "*back row is filled*"

ω_1 ω_2 ω_3



Each of these outcomes is in the event

$$E = \{\omega_1, \omega_2, \omega_3, ...\}$$

Events are sets of outcomes

List all events for a coin flip?

$$\Omega = \{H, T\}$$

Events are sets of outcomes

List all events for a coin flip?

$$\Omega = \{H, T\}$$

$$\mathcal{F} = \left\{ \{H, T\}, \{T\}, \{H\}, \underbrace{\varnothing}_{\text{empty set symbol}} \right\}$$

- Event *something* happens: contains every outcome
- Event *nothing* happens: $\varnothing$
- Event *heads*
- Event *tails*

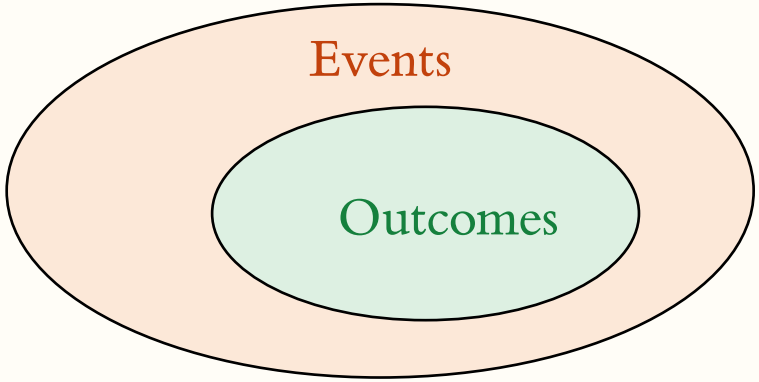
Events are sets of outcomes

List all events for a coin flip?

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- Event *something* happens: contains every outcome
- Event *nothing* happens: $\varnothing$
- Event *heads*
- Event *tails*



Every outcome is an event

Not every event is an outcome!

The event space $\mathcal{F}$ is the set of all possible events

Coin toss:

$\mathcal{F} = \{ \{H, T\}, \{T\}, \{H\}, \varnothing \}$

Each event is a set

So $\mathcal{F}$ is a set of sets!

The event space $\mathcal{F}$ is the set of all possible events

Coin toss:

$\mathcal{F} = \{ \{H, T\}, \{T\}, \{H\}, \varnothing \}$

Each event is a set

So $\mathcal{F}$ is a set of sets!

	Number of…	
	Outcomes	**Events**
Coin flip	2	$2^2 = 4$
Lecture theatre	2^{100}	$2^{2^{100}}$

Events have a set algebra

What is an algebra?

Rules for elements to interact with each other *and make babies*

Numbers

$$1 + 2 = \underbrace{3}_{\text{also a number!}}$$

$$1 \times 2 = 2$$

$\ldots$

Events have a set algebra

What is an algebra?

Rules for elements to interact with each other *and make babies*

Sets

Symbol	Example	Meaning
$\cup$	$X \cup Y$	Union (“*or*”)
$\cap$	$X \cap Y$	Intersection (“*and*”)
$\setminus$	$X \setminus Y$	Complement (“*without*”)

Events have a set algebra

Symbol	Example	Meaning
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Event X *Back row completely filled*
Event Y_1 *Front row completely filled*
Event Y_2 *Back row partially filled*

Express the following as English sentences

$X \cap Y_1$ $X \cap Y_2$

$X \cup Y_1$ $X \cup Y_2$

$X \setminus Y_1$ $X \setminus Y_2$

Each event has a probability

$$\underbrace{\text{Event } A}_{\text{back row filled}} \rightarrow \underbrace{0.3}_{\text{probability}}$$

Probability function

$$P : \mathcal{F} \rightarrow [0, 1]$$

- *"goes from set of events $\mathcal{F}$, to numbers between 0 and 1"*

What does it mean to have a probability?

Probability expresses partial knowledge of an event

With enough knowledge and insight, could I predict tomorrow's weather without uncertainty?

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Probability expresses partial knowledge of an event

With enough knowledge and insight, could I predict tomorrow's weather without uncertainty?

✗ correct probability

✓ Best probability given my knowledge of the world ← *subjective*!

Summary of probability objects

Probability space consists of three things

$$\left(\underbrace{\Omega}_{\text{Outcome space}}, \overbrace{\mathcal{F}}^{\text{event space}}, \underbrace{P}_{\text{probability function}} \right)$$

Summary of probability objects

Summary of probability objects

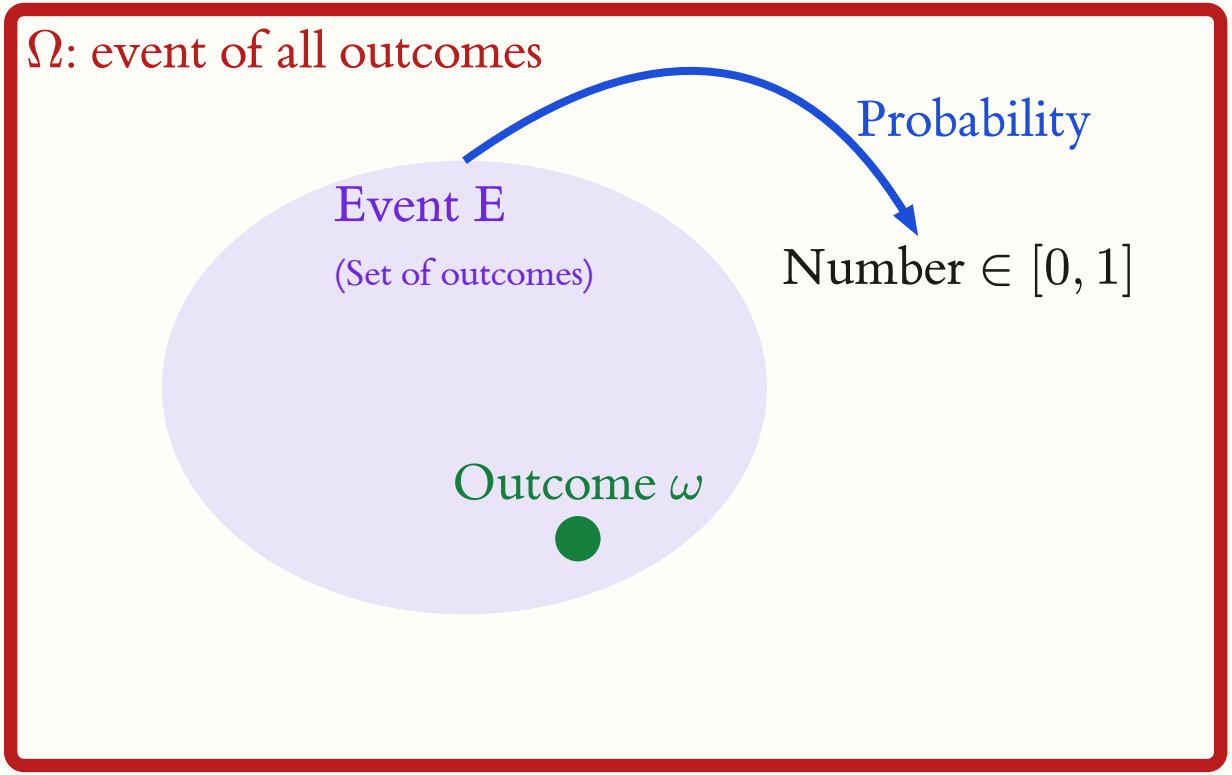
Probability space consists of three things

$$\left(\underbrace{\Omega}_{\text{Outcome space}}, \overbrace{\mathcal{F}}^{\text{event space}}, \underbrace{P}_{\text{probability function}} \right)$$

Summary of probability objects

Set	Description	Individual element
Ω	Every possible seating combination	$\omega \in \Omega$: *individual layout*
$\mathcal{F}$	Sets of seating combinations	$E \in \mathcal{F}$: *layouts where back row is filled*
P	Function	$P(E) = 0.3$ *ie 30%*

...graphically

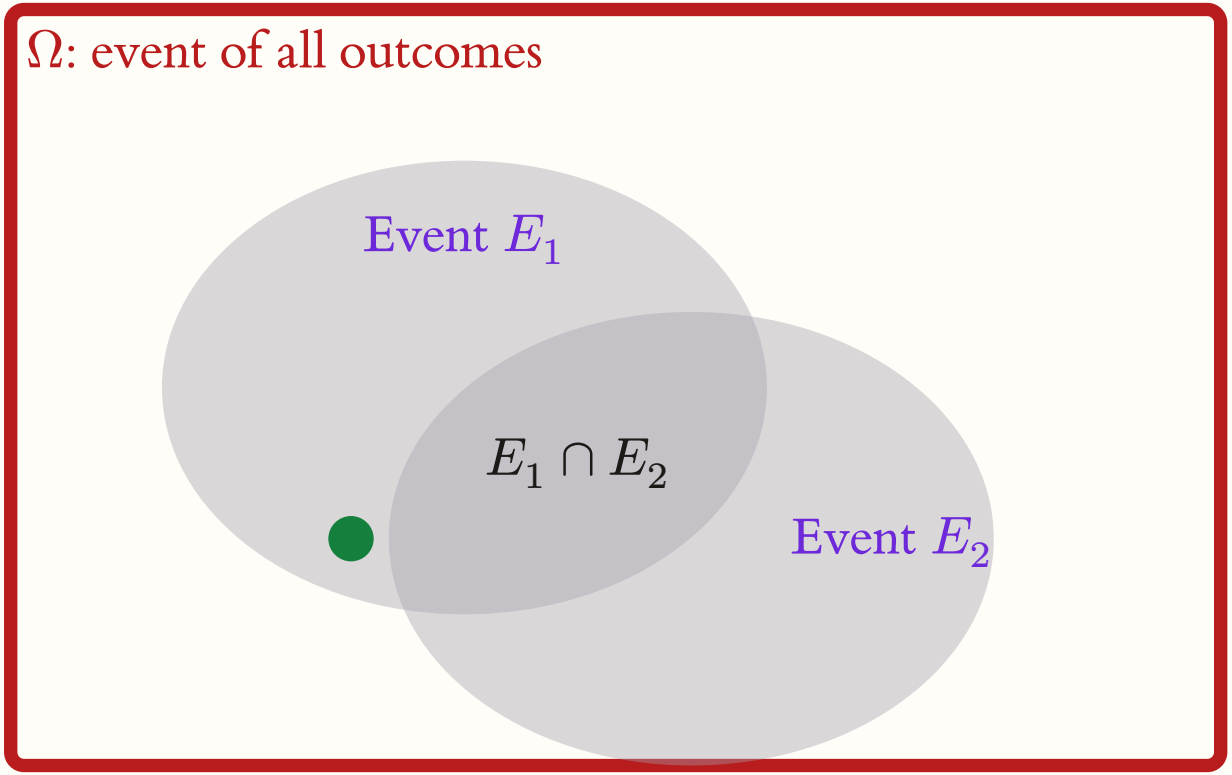


E : *"back row is filled"*

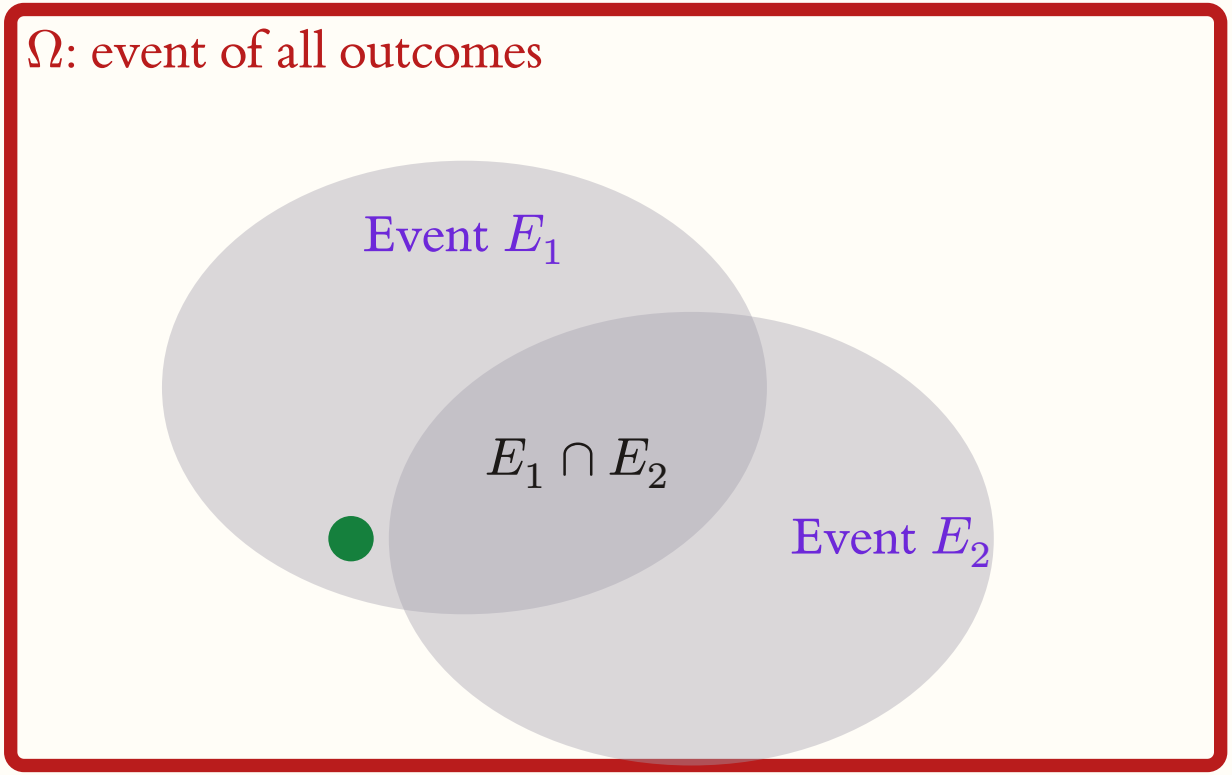
ω : *"single seating arrangement"*

$P(E)$: 0.3

Probability of union of events



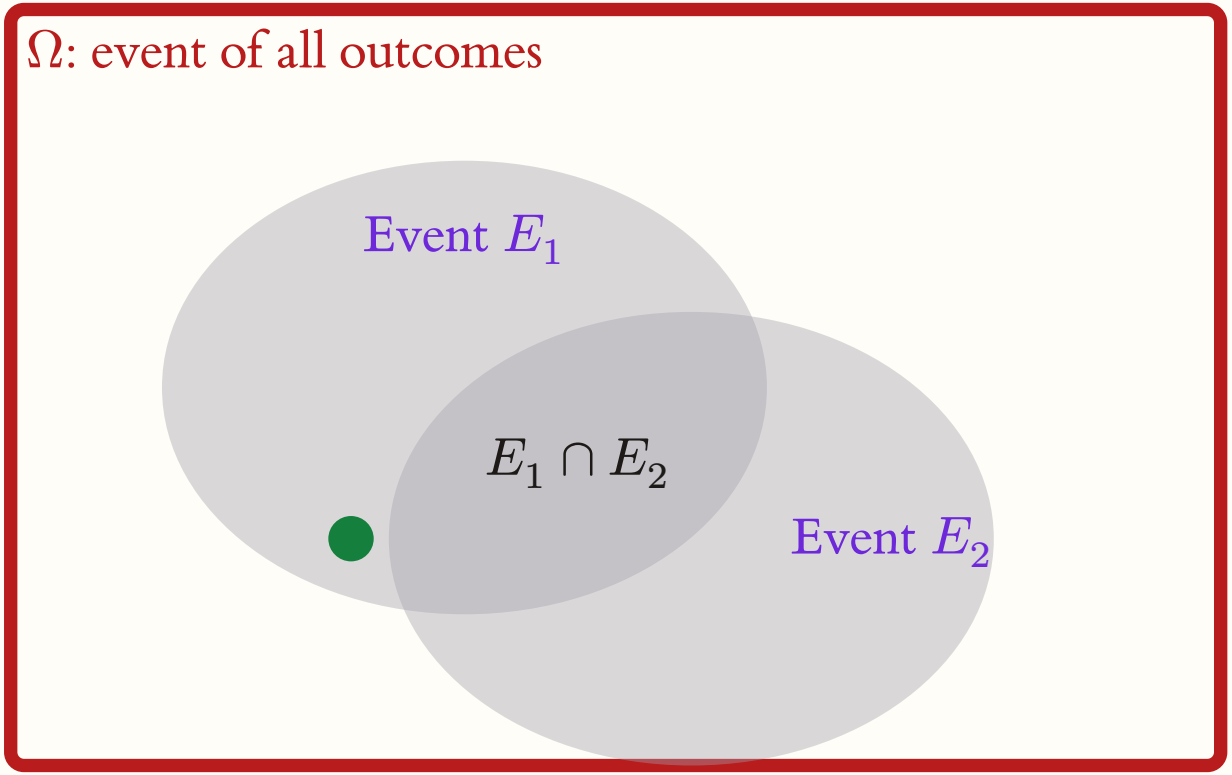
Probability of union of events



$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$P(\text{back row or front row filled?})$

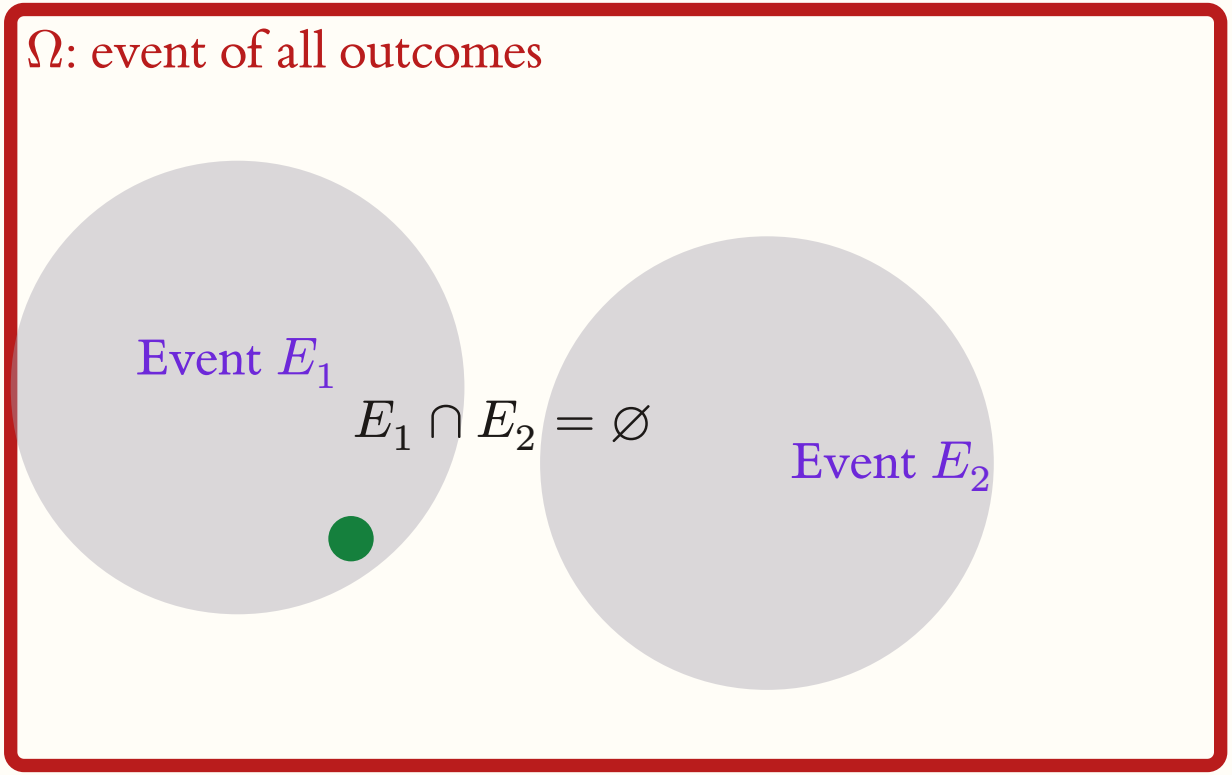
Probability of union of events



$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$P(\text{back row or front row filled?}) = P(\text{back}) + P(\text{front}) - P(\text{back and front})$$

Mutually exclusive events

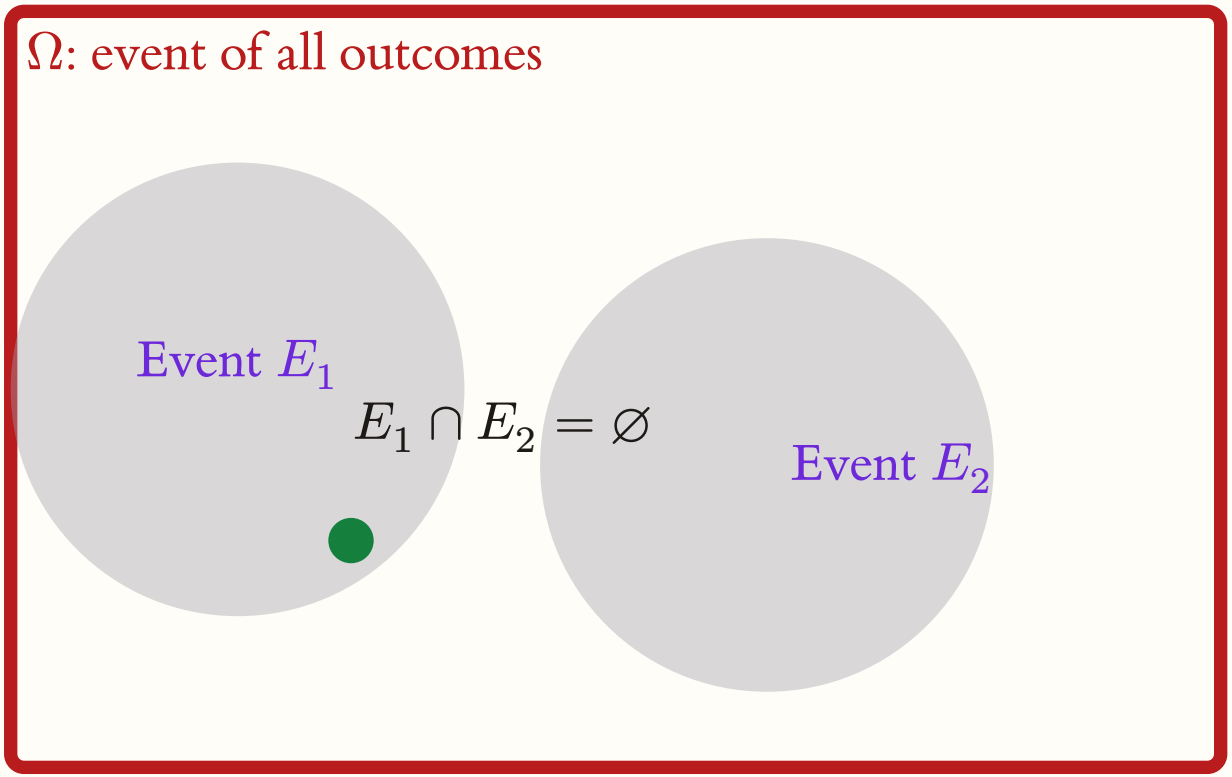


$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - \underbrace{P(E_1 \cap E_2)}_{=0}$$

Condition for mutually exclusive events

$$P(E_1 \cap E_2) = 0$$

Mutually exclusive events



Condition for mutually exclusive events

$$P(E_1 \cap E_2) = 0$$

Implication[1]

$$\underbrace{P(E_2 \mid E_1)}_{\text{E2 'given' E1}} = P(E_1 \mid E_2) = 0$$

[1] Vertical lines are called conditional probabilities

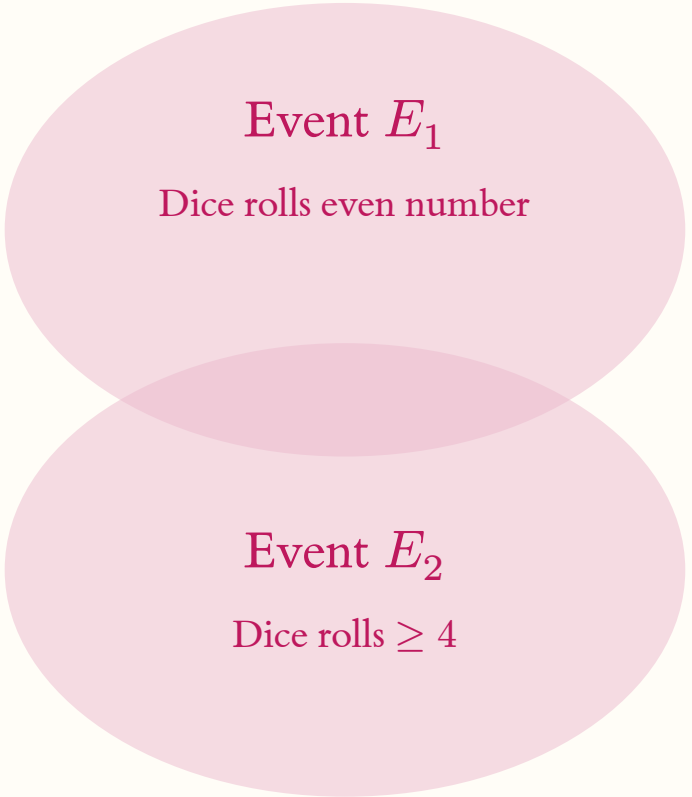
Independent events are not mutually exclusive

Definition: independent events

Knowledge of one event *does not change* probability of the other

- $P(E_1 \mid E_2) = P(E_1)$
- $P(E_2 \mid E_1) = P(E_2)$

Independent?



Independent events are not mutually exclusive

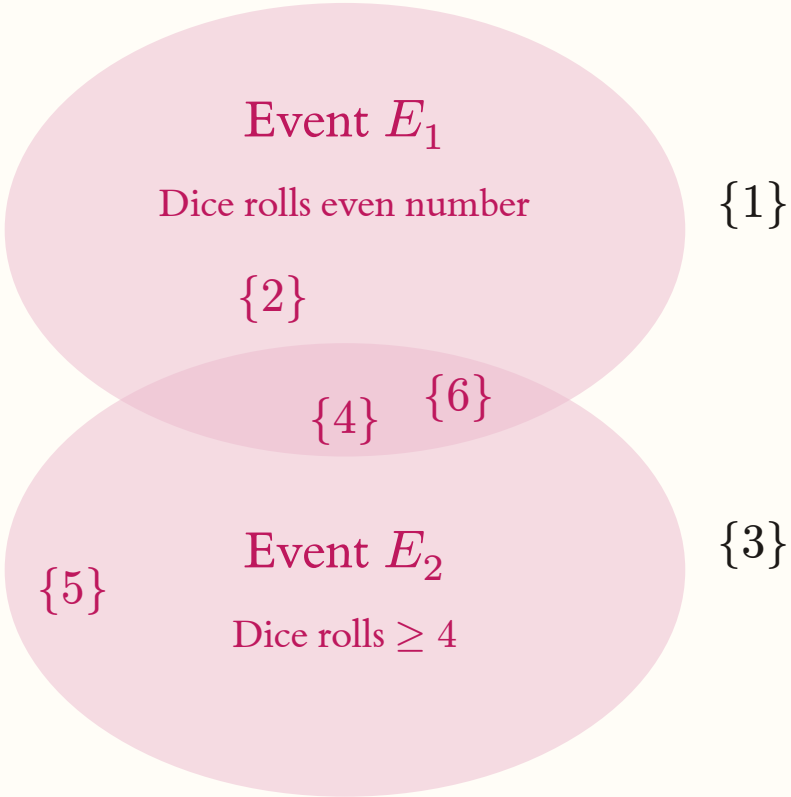
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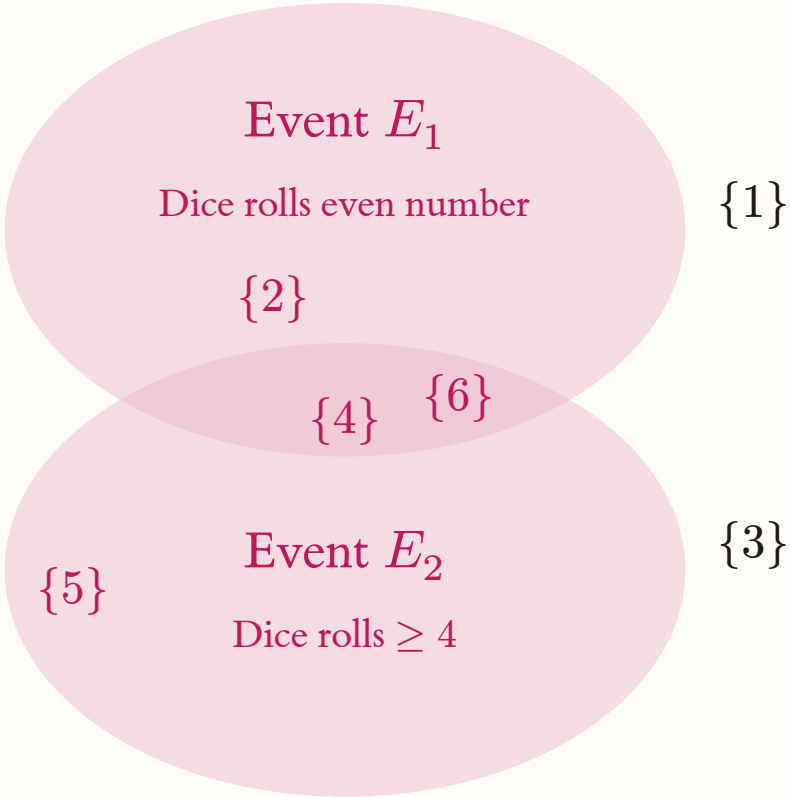
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$$P(E_2 \mid E_1) = \frac{2}{3} \neq P(E_2)$$ ✗

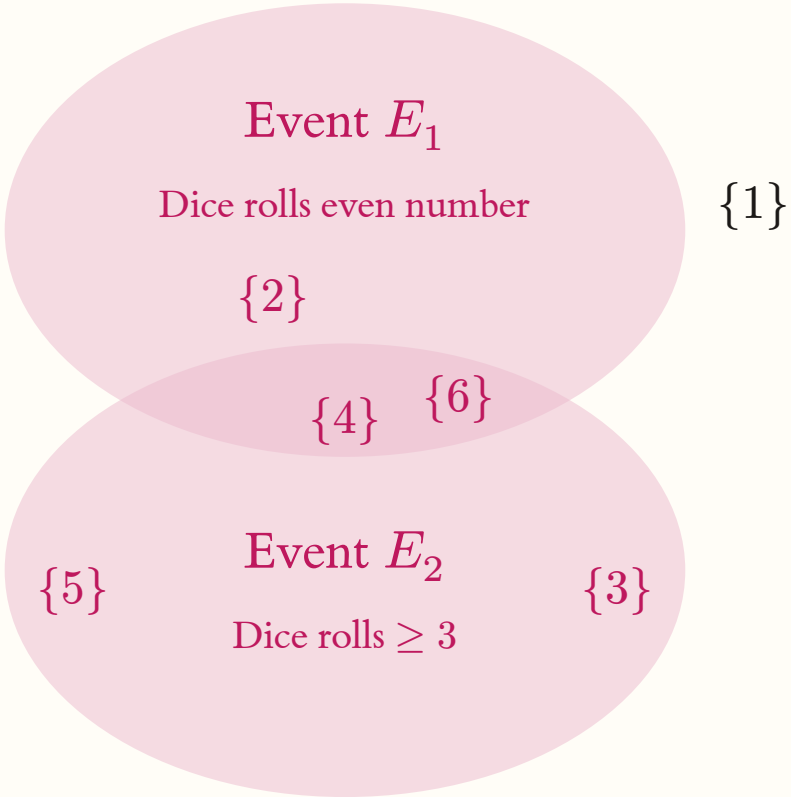
Independent?



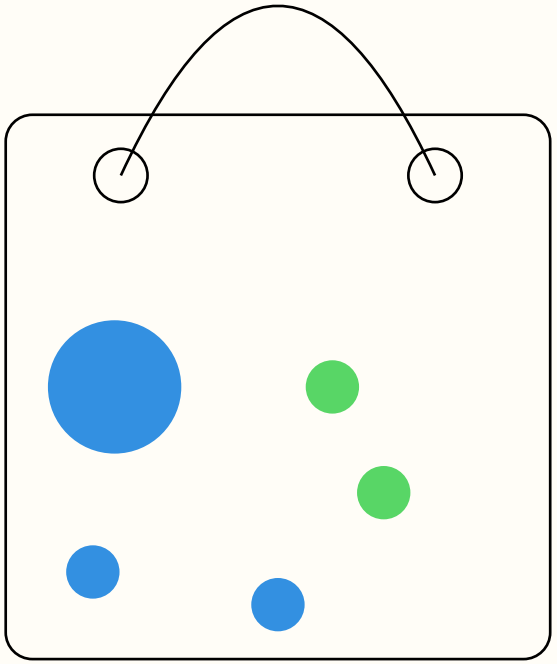
Independent events are not mutually exclusive

Independent?

(Do at home)



Independent? Mutually exclusive?



Experiment:

Take two balls from bag

E_1 : First ball is blue

E_2 : Second ball is green

E_3 : First ball is big

Questions

$E_3 \subset E_1$?

$E_2 \subset E_1$?

$P(E_2 \mid E_1)$?

$P(E_3 \mid E_1)$?

$E_1^{\complement} \cap E_3$?

$P(E_3 \cup E_1)$?

Independent? Mutually exclusive?

Terminology

A^C : "not A" (complement of A)

$\subset$: "subset of"

Random variables[1]

[1] neither random, nor variables

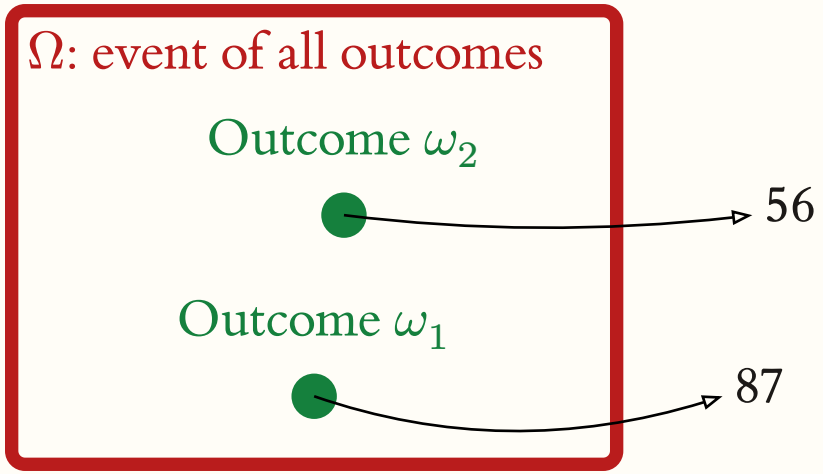
What are random variables?

They are

Quantitative questions about the experiment

Functions that map from outcomes to numbers

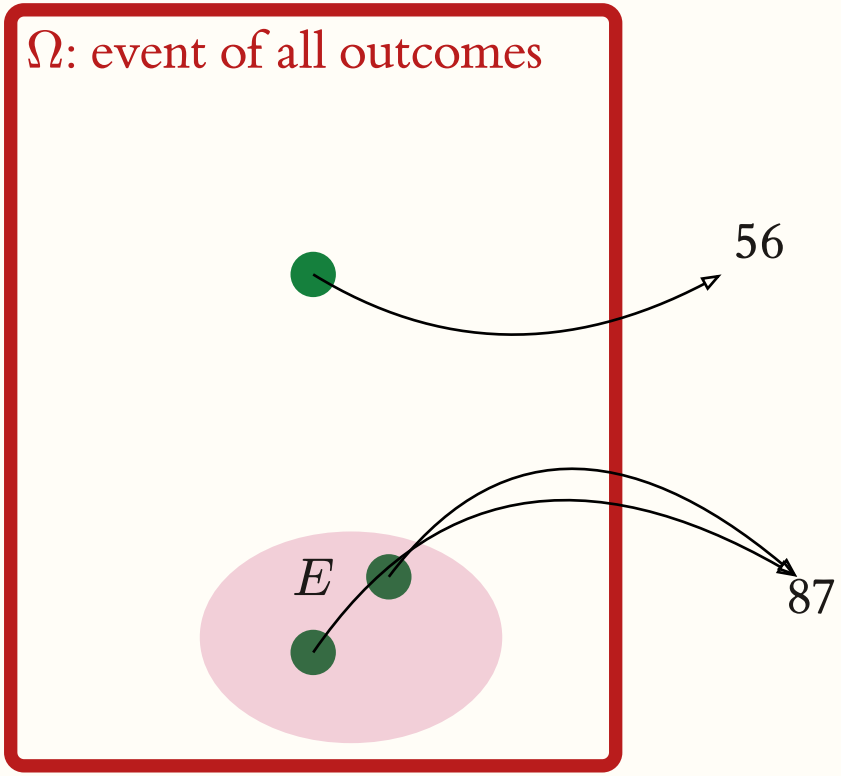
X: What was the number of unfilled seats?



- $X(\omega_1) = 87$
- $X(\omega_2) = 56$

Random variable taking particular value is event

"There were 87 unfilled seats"



$$E = \{\omega \in \Omega : X(\omega) = 87\}$$

Probability of 87 unfilled seats?

- $P(E) = P(X(\omega) = 87)$

Flipping a coin

Random variable X:

Heads $\Rightarrow$ 8

Tails $\Rightarrow$ 0

Random variables are not random. *The outcome is*.

Random variables are the deterministic mapping from outcomes to numbers

Support of a random variable

Random variable X:

Heads $\Rightarrow$ 8

Tails $\Rightarrow$ 0

$\text{supp}(X) = \{0, 8\}$

Support

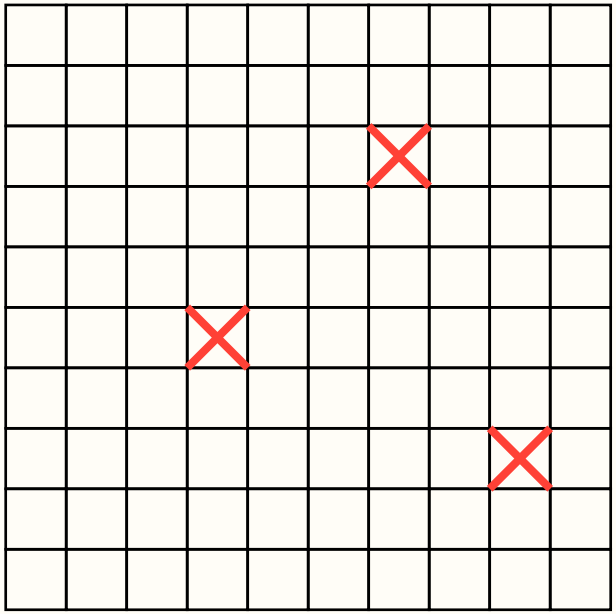
Set of possible values an RV can take

$=$

Smallest set A such that $P(X \in A) = 1$

Support of a random variable

(10 × 10 theatre)



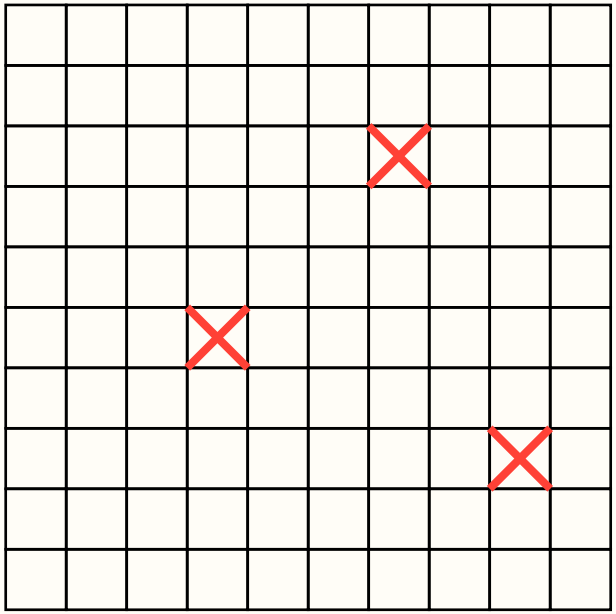
= empty seat

X : number of unfilled seats

Support of X?

Support of a random variable

(10×10 theatre)



= empty seat

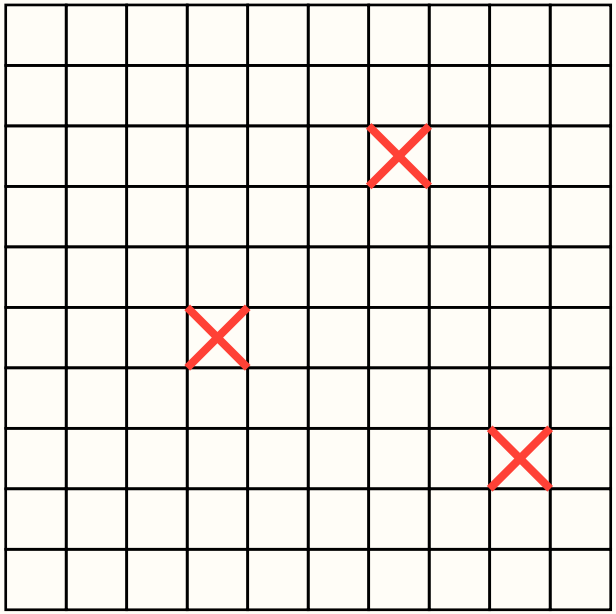
X : number of unfilled seats

Support of X?

$$\begin{aligned} \text{supp}(X) &= \{0, 1, 2, ..., 100\} \\ &= [0, 100] \cap \mathbb{N} \end{aligned}$$

Support of a random variable

$(10 \times 10$ theatre)



= empty seat

Y : is the back row filled?

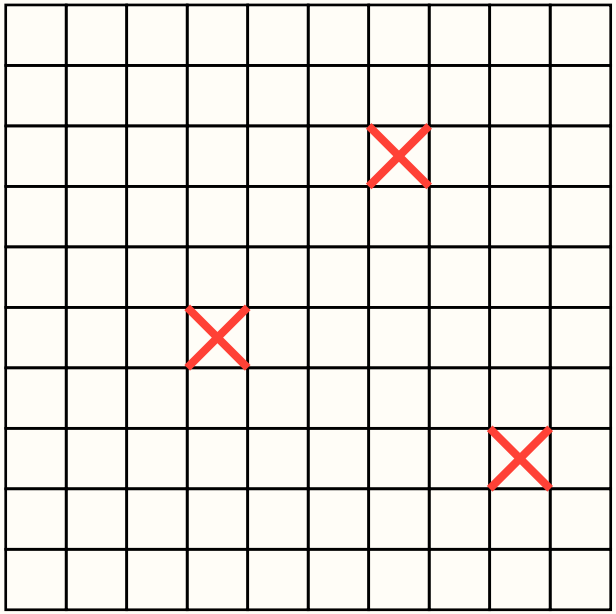
Z : height of first person on third row

$\text{supp}(Y)$

$\text{supp}(Z)$?

Support of a random variable

$(10 \times 10$ theatre)



= empty seat

Y : is the back row filled?

Z : height of first person on third row

$\text{supp}(Y)$	$\{0, 1\}$ or $\{\text{true}, \text{false}\}$
$\text{supp}(Z)$?	e.g. $(130, 210)$

Two flavours of random variable

Discrete random variables

- Support is countable set

X : *How many unfilled seats?*
Y : *Is back row filled?*

Continuous random variables

- Support is continuous range of values

Z : *Height of particular student?*

Two flavours of random variable

Discrete random variables

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Countable sets

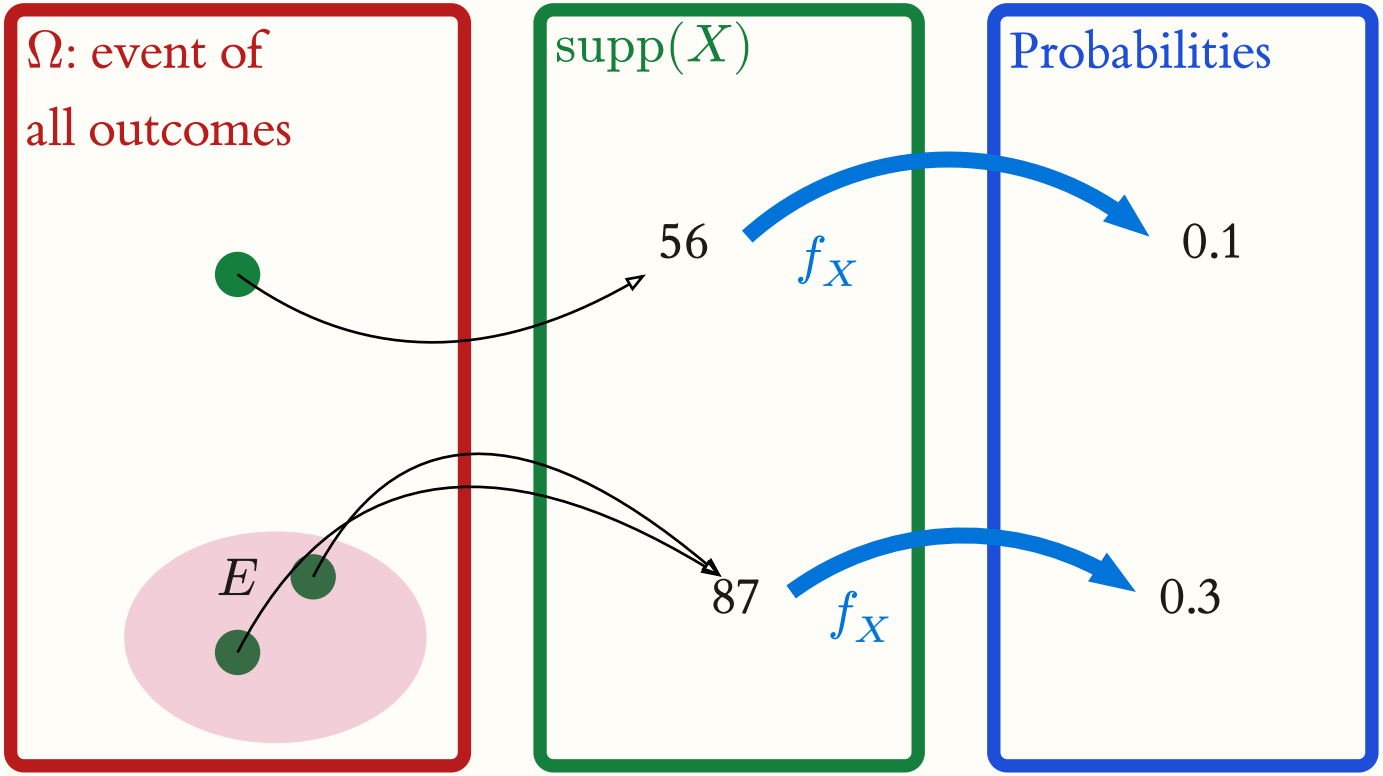
- $\text{supp}(\text{X}) = \{0, 1\}$
- $\text{supp}(\text{Y}) = \{0, 1, 2, ...100\}$

Uncountable sets

- $\text{supp}(\text{Z}) = (130, 210)$

Discrete random variables have probability mass functions

X : *How many unfilled seats?*

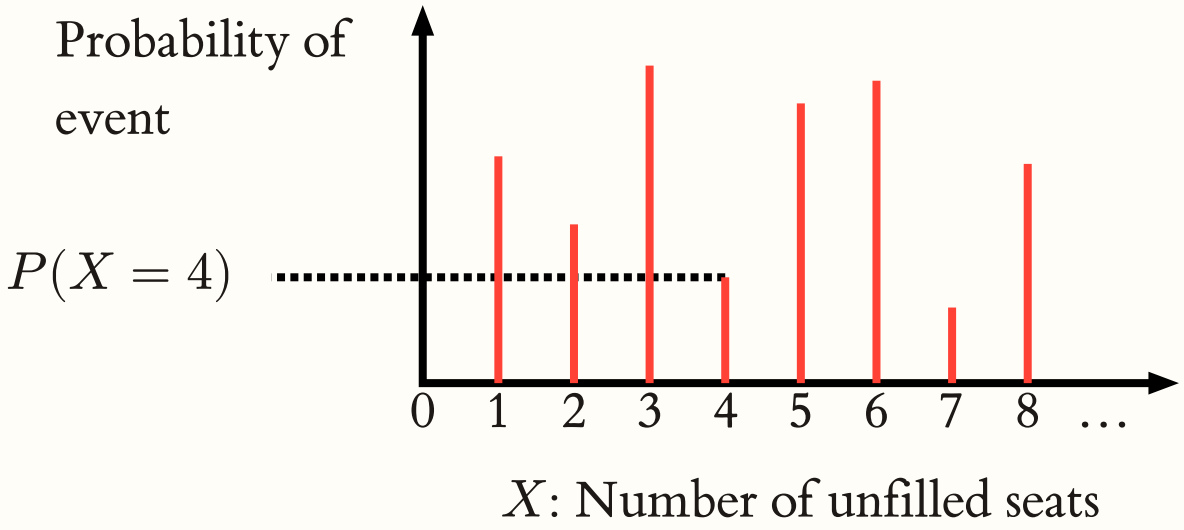


PMF f_X:

- Maps *value taken by* RV to corresponding *probability*

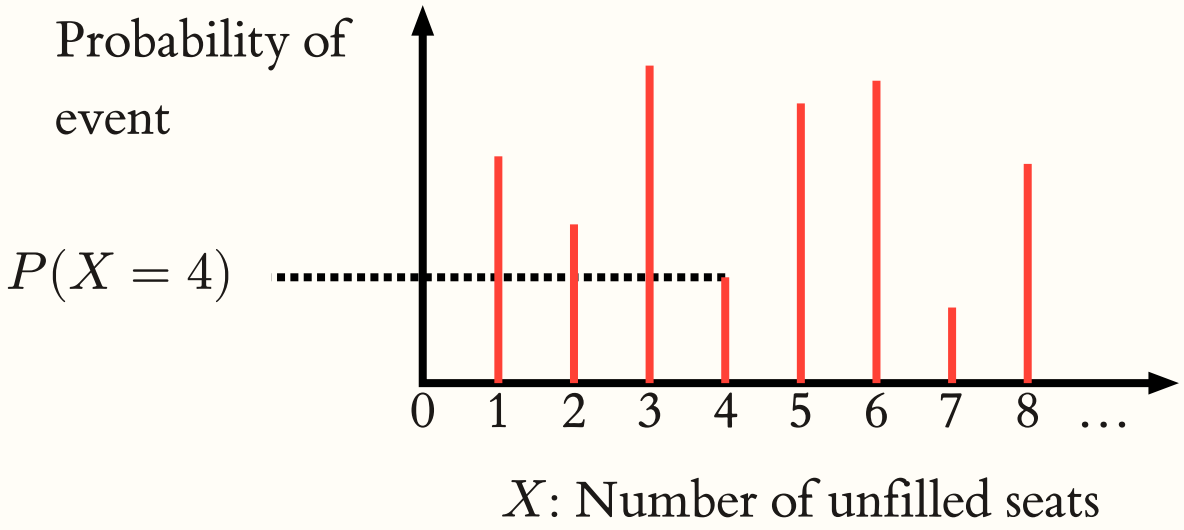
- $f_X(56) = 0.1$
- $f_X(87) = 0.3$

Graph of probability mass function



Total length of all red lines?

Graph of probability mass function



Total length of all red lines?

$$P(X \in \text{supp}(X)) = 1$$

Recurring flavours of random variables across different experiments

Bernoulli Random variables

Reminder

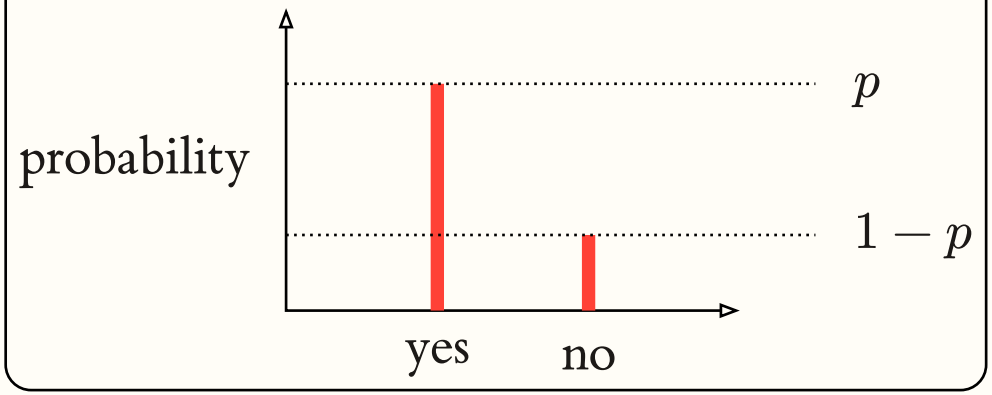
Random variables are quantitative questions about the experiment

Bernoulli RVs

- answer is yes/no (1/0)
- *"Is the back row filled"*?

Maths	$X \sim \text{Bern}(p)$
English	X *is distributed as*[1] a Bernoulli RV, with probability p (of yes)

PMF of Bernoulli



[1] $\sim$: "*is distributed as*"

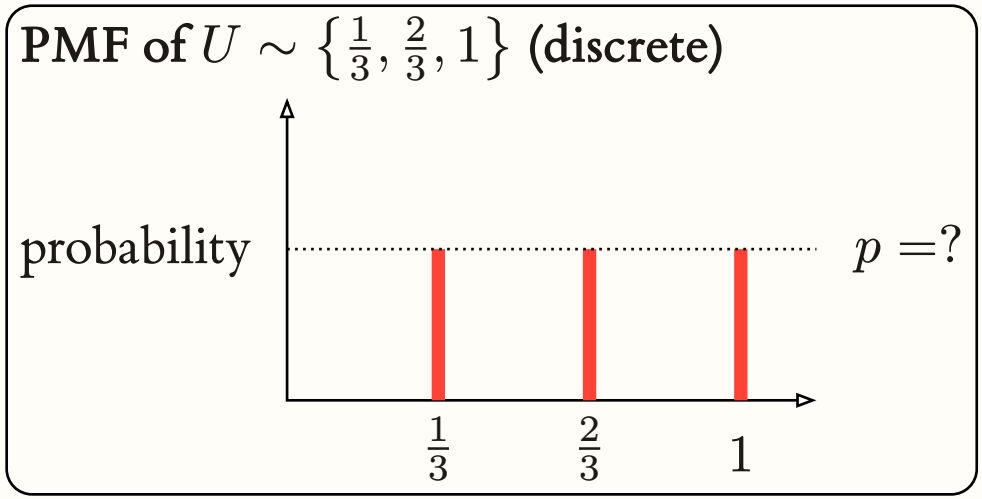
Uniform random variables

Maths $X \sim \mathrm{U}(p)$

English X *is distributed as* a Uniform RV, on the set S

Probability of *every* value in support is equal[1]

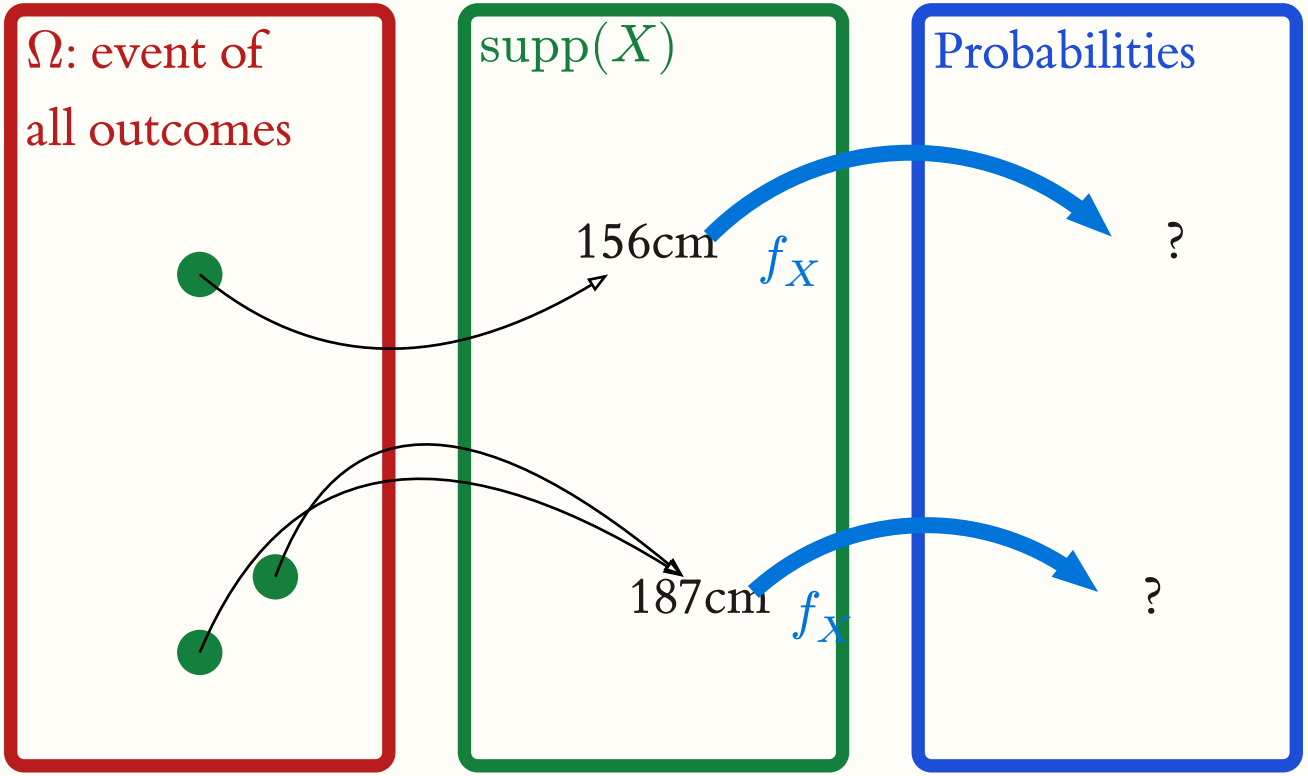
Discrete? Continuous? Depends on support!



[1] Likely means we know little about the experiment

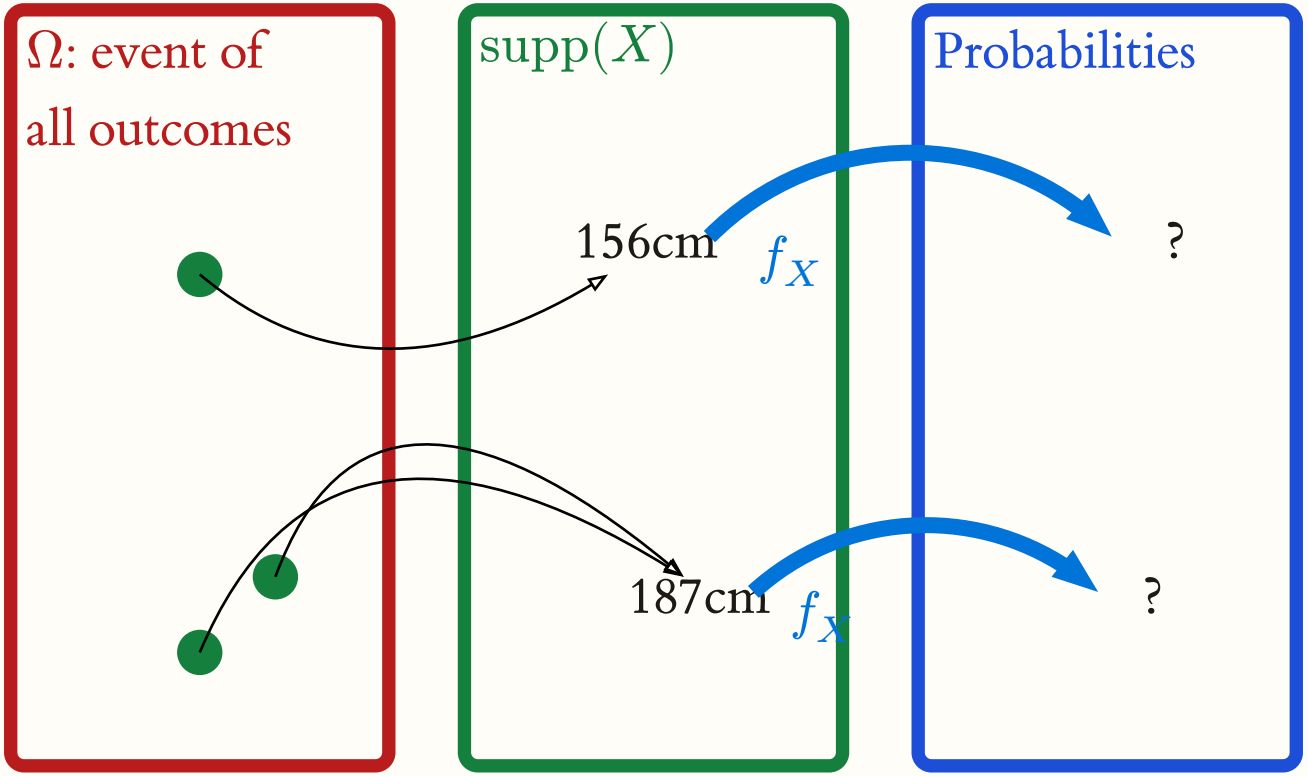
Probability density functions for continuous random variables

Height of a person?



Probability density functions for continuous random variables

Height of a person?

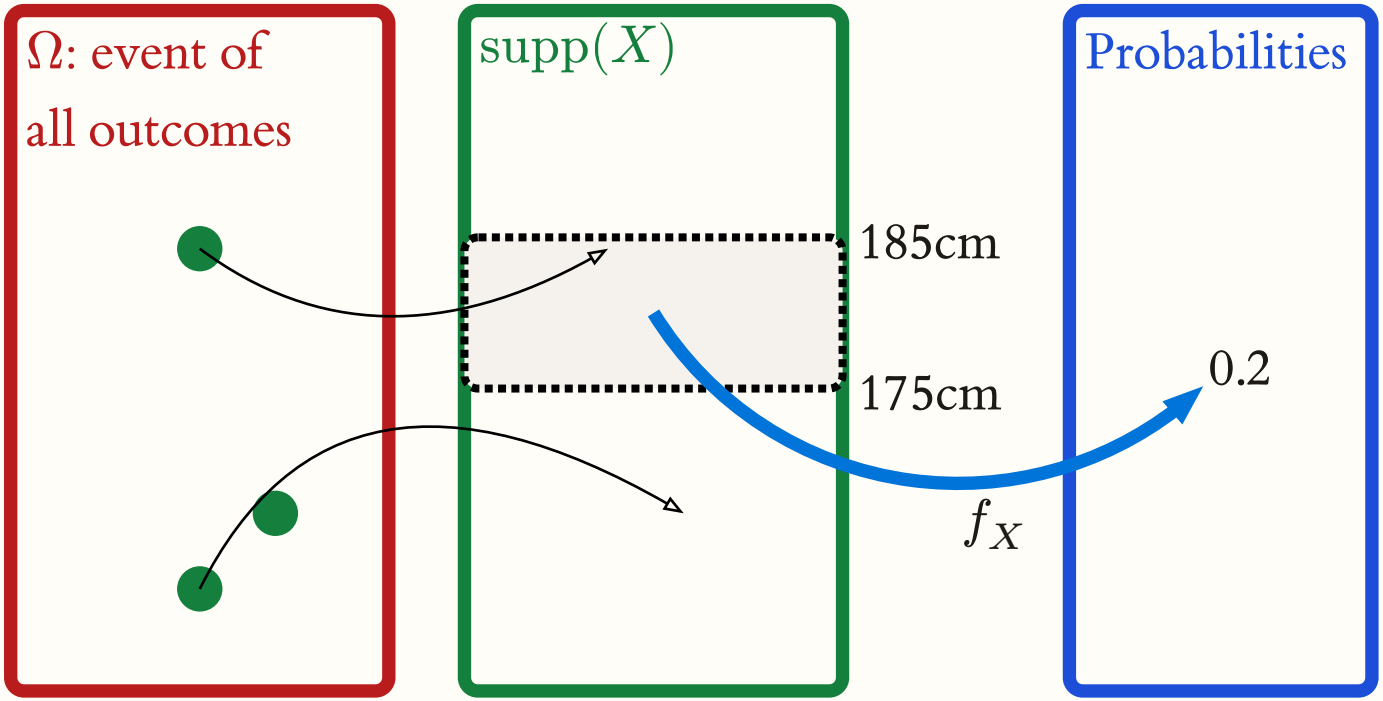


$$\begin{aligned} & P(X = 156\text{cm}) \\ = & P(X = 156.00000...\text{cm}) \\ = & ...0? \end{aligned}$$

Houston, we have a problem!

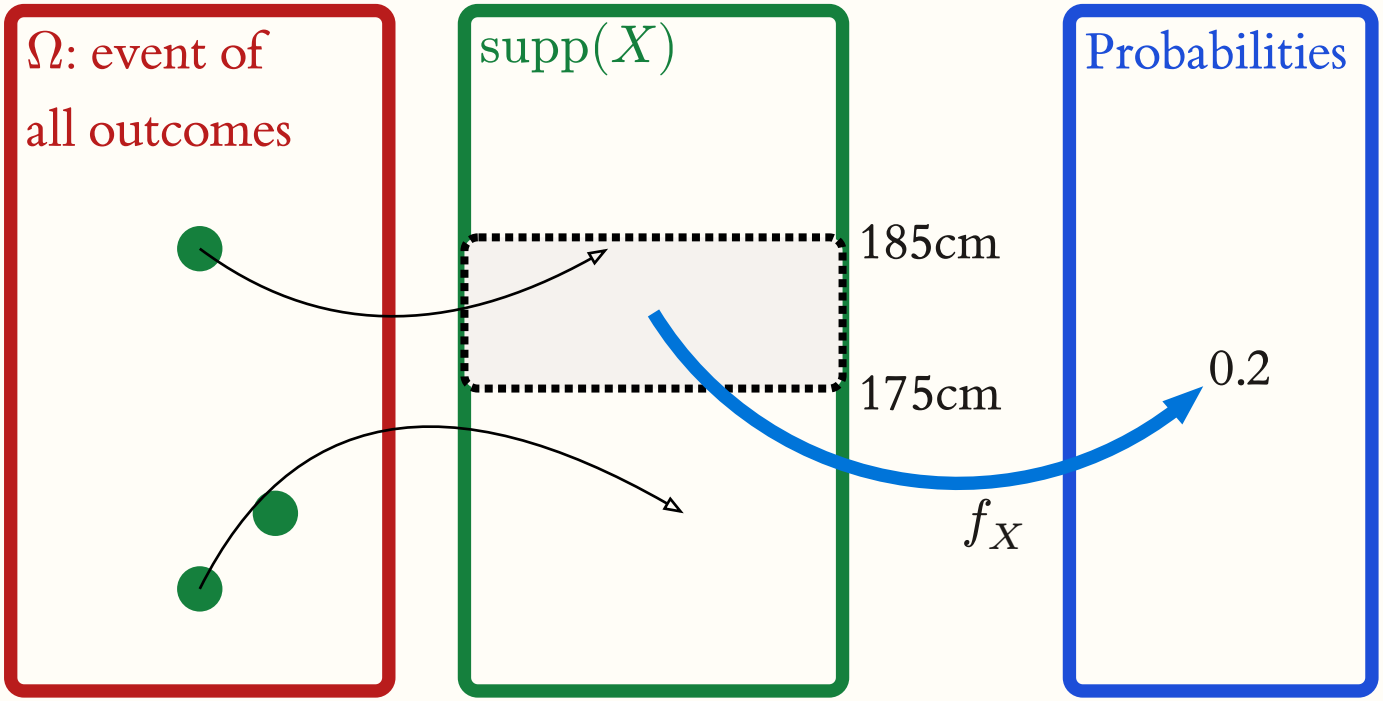
PDFs provide probabilities for ranges of outcomes

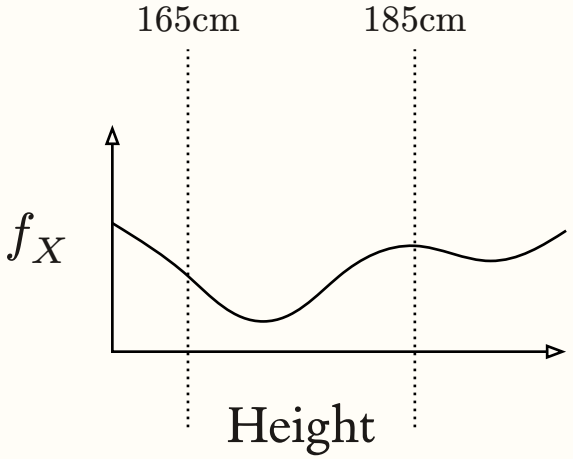
X : Height of a person



PDFs provide probabilities for ranges of outcomes

X : Height of a person





Integral is area under curve

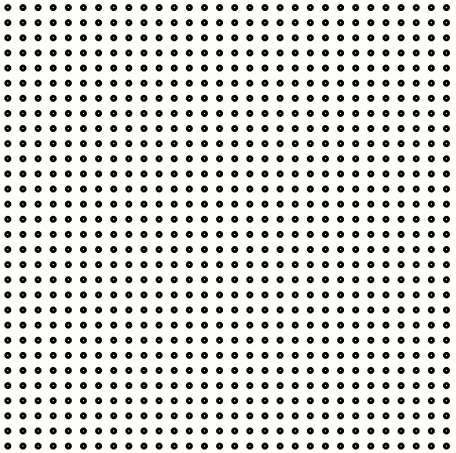
$$P(165 \leq X \leq 185) = \int_{165}^{185} f_X(x)\ \mathrm{d}x$$

$$P(a \leq X \leq b) = \int_a^b f_X(x)\ \mathrm{d}x$$

Analogy

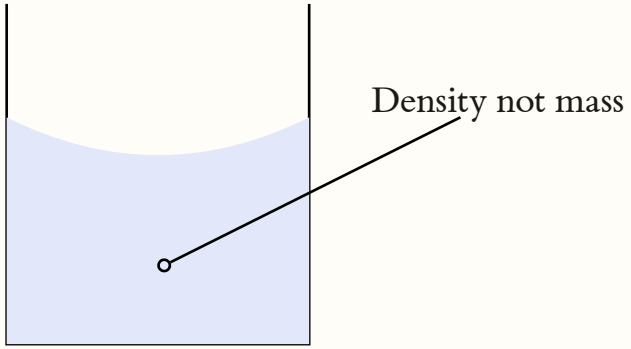
Grains of sand are countable (finite)

- Each grain has a nonzero *mass*



Water (continuum) occupies a range of space

- Can't measure water *mass* at a point
- Can measure water *density* at a point



- mass of a *volume* (3D range) of water makes sense

$$\text{mass} = \int_{\text{volume}} \text{density} \; \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$$

Summary

Experiments have outcomes

Sets of outcomes are events

Random variables map outcomes to numbers

Random variable taking particular value is an event

PMFs/PDFs give probabilities of such events

PDF flavours again

- Random variables ask quant questions
- Certain flavours of question are common

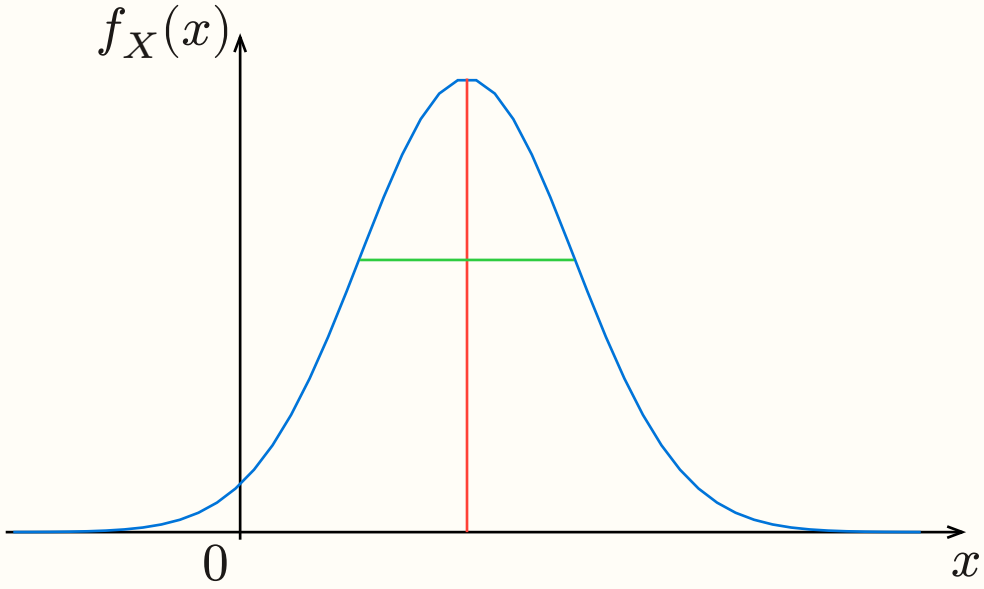
Name	Question	Notation
Bernoulli	yes/no	$\text{Bern}(p)$
Uniform	know nothing	$U(S)$
Gaussian[1]	???	???



[1] Galton board

The most important flavour of PDF

Gaussian distribution![1]

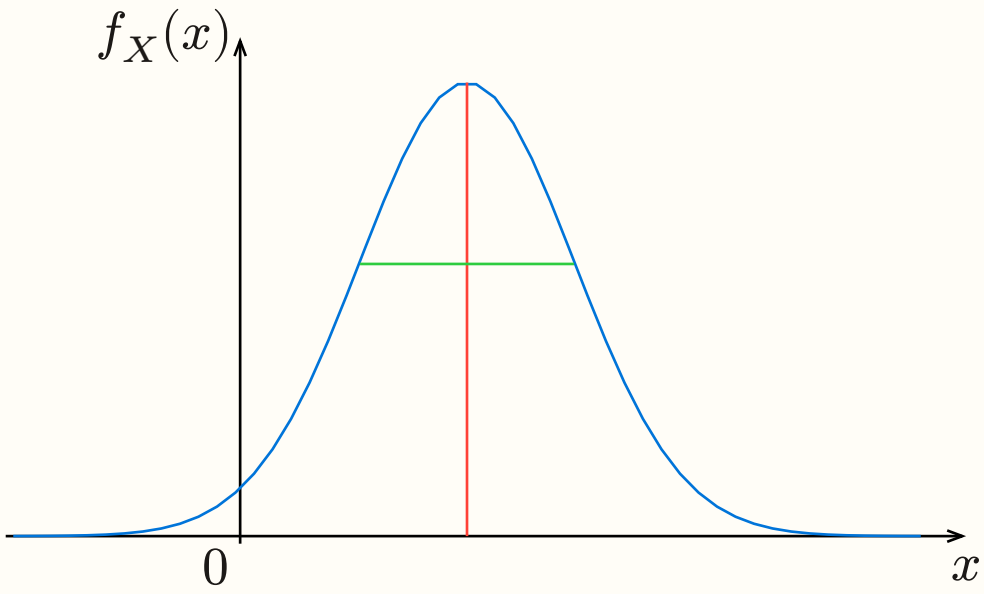


$$X \sim \mathcal{N}(\mu, \sigma^2)$$

[1] also known as Normal distribution or bell curve

The most important flavour of PDF

Gaussian distribution![1]



$$X \sim \mathcal{N}(\mu, \sigma^2)$$

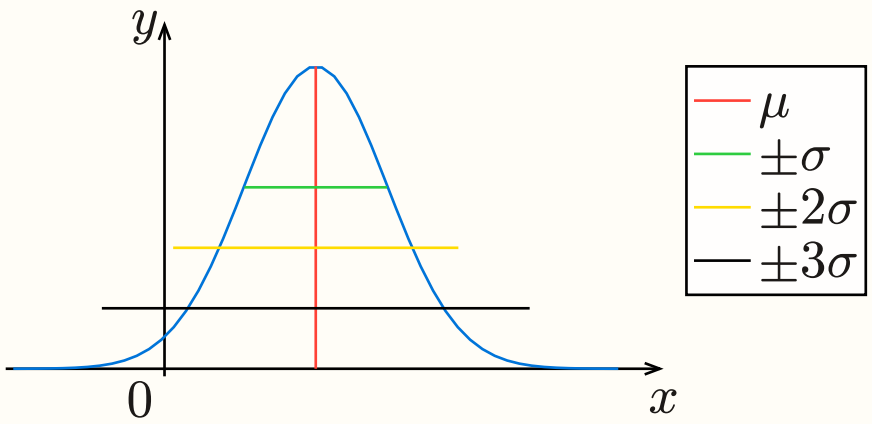
μ (mean)
$\pm\sigma$ (standard deviation)

- Occurs everywhere! Why?
- $X \sim \mathcal{N}(\mu, \sigma^2)$
- "*X is distributed as a Gaussian with mean μ and variance σ^2*"

Standard deviation (σ) is square root of variance (σ^2)

[1] also known as Normal distribution or bell curve

The most important flavour of PDF



Important table

Number of stds	Probability within
$\pm\sigma$	0.68
$\pm 2\sigma$	0.95
$\pm 3\sigma$	0.997

Regardless of actual μ, σ

Statistics

Statistics is extrapolating from limited data

Experiment

- Measure 100 people's heights at Sussex
- Average was 172cm

- Which claim do you bet on?

Claim A

- Average Sussex uni height is inside 172 ± 2.4cm

Claim B

- Average UK height is inside 172 ± 2.4cm

Statistics is extrapolating from limited data

Experiment

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Claim B

- Average UK height is inside 172 ± 2.4cm

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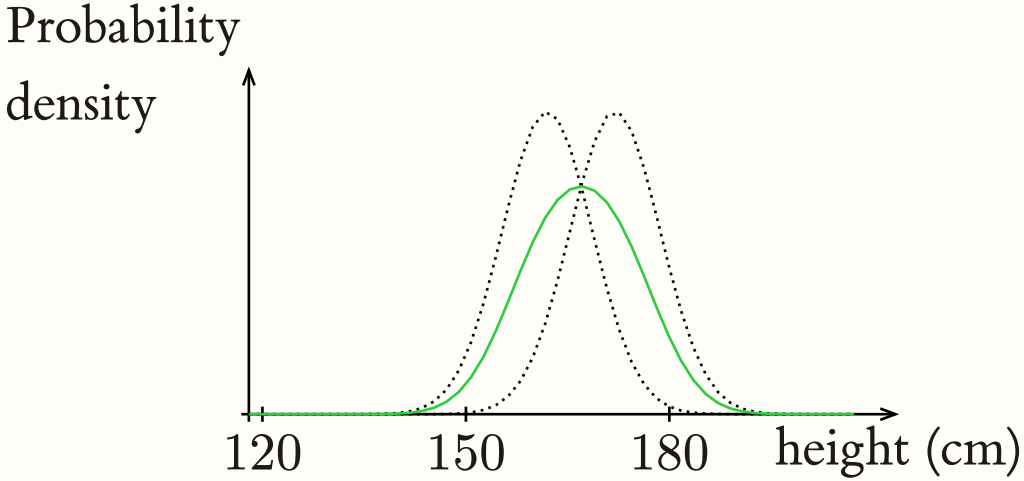
Common assumption

Unmeasured behaves like measured

Not equally safe

What this course is about!

True[1] distribution[2] of Sussex uni heights



How would we summarise this PDF?

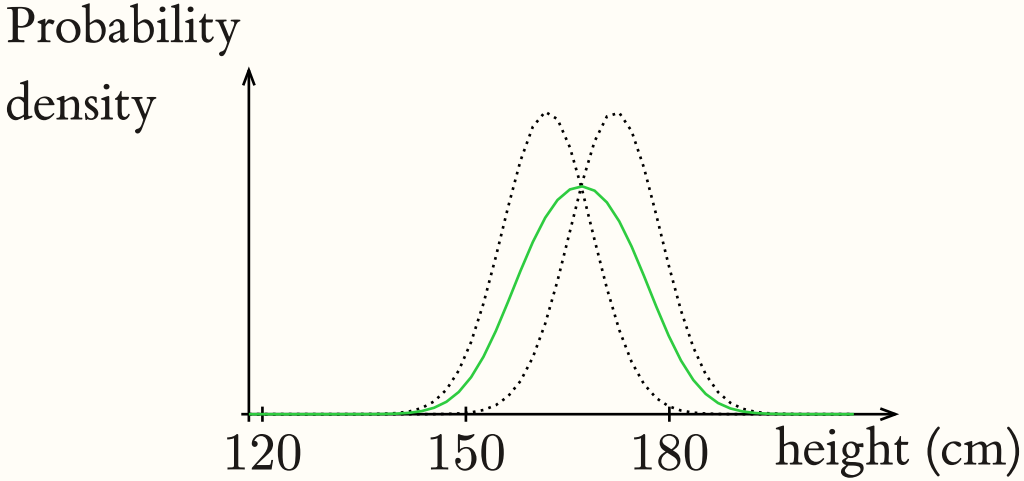
Middle? Spread? Shape?

[1] We'd never really have access to this

[2] i.e. PDF

Statistics in an unattainable omniscient world

True[1] distribution[2] of Sussex uni heights



How would we summarise this PDF?

Middle? Spread? Shape?

Population statistics

- Summaries of a PDF
- Fixed by universe.
- Never known by modeller

[1] We'd never really have access to this

[2] i.e. PDF

Population statistic I: expectation/mean

X : value of (fair) dice roll

1	2	3	4	5	6

One type of "middle" for a distribution

Formula[1] (discrete RV)

$$\mathbb{E}[X] = \sum_{x \in \text{ supp}(X)} x \times f_X(x)$$

[1] Learn to read/understand this maths notation!

Population statistic I: expectation/mean

One type of "middle"
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X : value of (fair) dice roll

1	2	3	4	5	6

$$\mathbb{E}[X] = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + \ldots + 6 \times \frac{1}{6}$$

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$$= \sum_{i=1}^{6} i \times f_X(i) = 3.5 \quad \text{(unattainable)}$$

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$$= \sum_{i=1}^{6} i \times f_X(i) = 3.5 \quad \text{(unattainable)}$$

Never know true PDF:

- Is your die truly fair?

[1] Learn to read/understand this maths notation!

Expectation is linear

- X : number rolled on die
- $Y = X^2$
- $Z = 4X + 6Y$

1	2	3	4	5	6

- $\mathbb{E}[X] = 3.5$
- $\mathbb{E}[Y] = \sum_{i=1}^{6} i^2 f_Y(i) = 15\frac{1}{6}$[1]

$$E[Y] = (E[X])^2?$$

[1] Verify at home. Good practice.

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- No. Can't square expectations. $3.5^2 = 12.25$

However

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$$E[Y] = (E[X])^2?$$

- No. Can't square expectations.
  $3.5^2 = 12.25$

However

$$\mathbb{E}[Z] = 4\mathbb{E}[X] + 6\mathbb{E}[Y] = 105 \quad \checkmark$$

[1] Verify at home. Good practice.

Expectation is linear

- X : number rolled on die
- $Y = X^2$
- $Z = 4X + 6Y$

1	2	3	4	5	6

- $\mathbb{E}[X] = 3.5 \quad \mathbb{E}[Y] = 15\frac{1}{6}$

Linearity definition[1]

Function f is linear if

$$f(ax + by) = af(x) + bf(y),$$

for constants $a, b \in \mathbb{R}$ *and variables* x, y

- $\mathbb{E}[Z] = \mathbb{E}[4X + 6Y] = 4\underbrace{\mathbb{E}[X] + 6\mathbb{E}[Y]}_{\text{linearity}}$

[1] Needs thought and practice. See lab/cheatsheet

Expectation is linear

Careful

Always (linearity):

- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

If independent:

- $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$[1]

Linearity definition

Function f is linear if

$$f(ax + by) = af(x) + bf(y),$$

for constants $a, b \in \mathbb{R}$ *and variables* x, y

- $\mathbb{E}[Z] = \mathbb{E}[4X + 6Y] = 4\underbrace{\mathbb{E}[X] + 6\mathbb{E}[Y]}_{\text{linearity}}$

[1] $Y = X^2$ is as dependent as it gets!

Population statistic II: median

Mean is vulnerable to outliers:

1	2	3	4	5	600

$$\begin{aligned}\mathbb{E}[X] &= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + ...5 \times \frac{1}{6} + 600 \times \frac{1}{6} \\ &= 102.5\end{aligned}$$

[1] i.e. expectation/mean

Population statistic II: median

Mean is vulnerable to outliers:

1	2	3	4	5	600

$$\mathbb{E}[X] = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + ...5 \times \frac{1}{6} + 600 \times \frac{1}{6}$$
$$= 102.5$$

Median

$P(X \geq \text{median}(X)) =$
$P(X \leq \text{median}(X))$

- Value? Unique? Vulnerable?

[1] i.e. expectation/mean

Population statistic II: median

Mean is vulnerable to outliers:

1	2	3	4	5	600

$$\mathbb{E}[X] = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + ...5 \times \frac{1}{6} + 600 \times \frac{1}{6}$$
$$= 102.5$$

Median

$P(X \geq \text{median}(X)) =$
$P(X \leq \text{median}(X))$

- Value? Unique? Vulnerable?

Most people earn less than the average[1] salary.

- True/false? Why?

[1] i.e. expectation/mean

Percentiles and interquartile range

50th percentile (median)

$$P(X < \text{median}) = \frac{50}{100} = 0.5$$

[1] Q3 is third quartile i.e. 75th percentile

Percentiles and interquartile range

50th percentile (median)

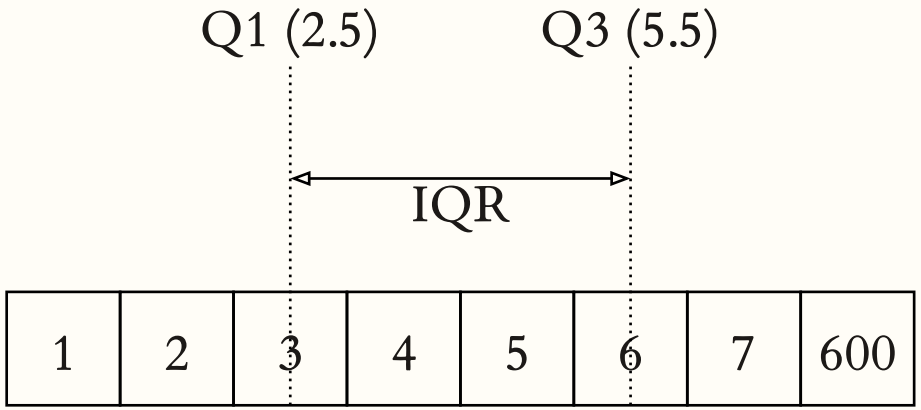
$$P(X < \text{median}) = \frac{50}{100} = 0.5$$

25th percentile (first quartile)[1]

$$P(X < \text{Q1}) = \frac{25}{100} = 0.25$$

pth percentile

$$P(X < v_p) = 0.01p$$



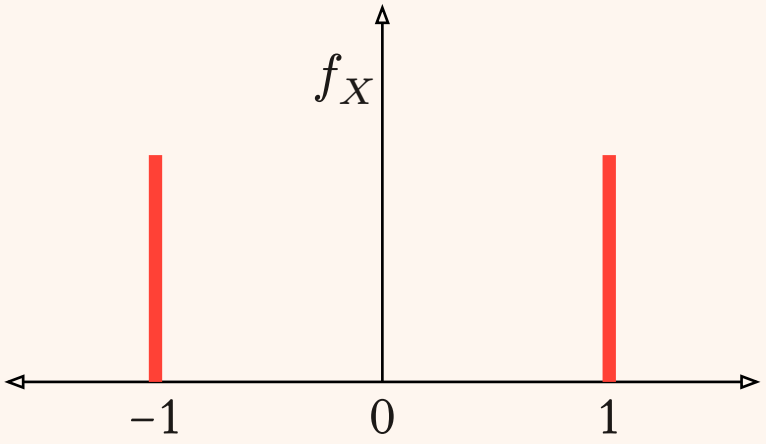
Interquartile range is a robust alternative to variance

[1] Q3 is third quartile i.e. 75th percentile

Population statistic III: variance

Goal: measure *spread* – average distance from mean

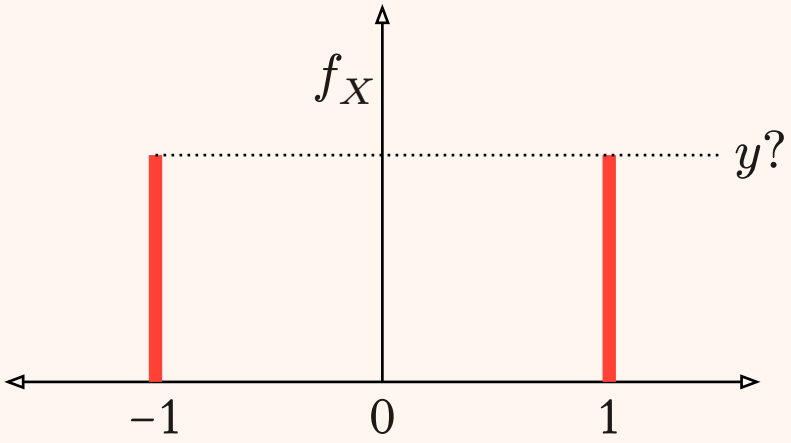
A toy distribution



Population statistic III: variance

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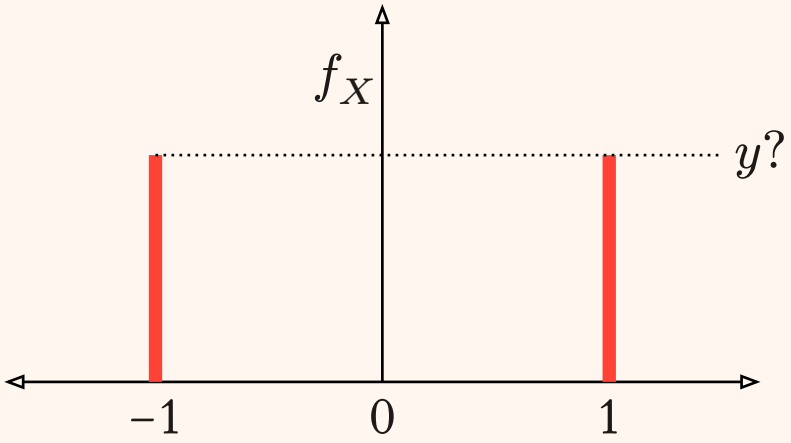
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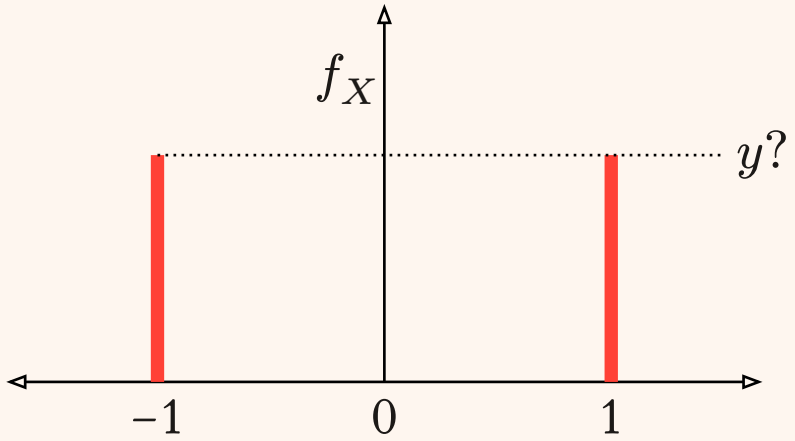


Try 1: $\mathbb{E}[(X - \mathbb{E}[X])?$

Population statistic III: variance

Goal: measure *spread* – average distance from mean

A toy distribution



Try 1: $\mathbb{E}[(X - \mathbb{E}[X])$?

- Always zero: prove at home (linearity)

Variance

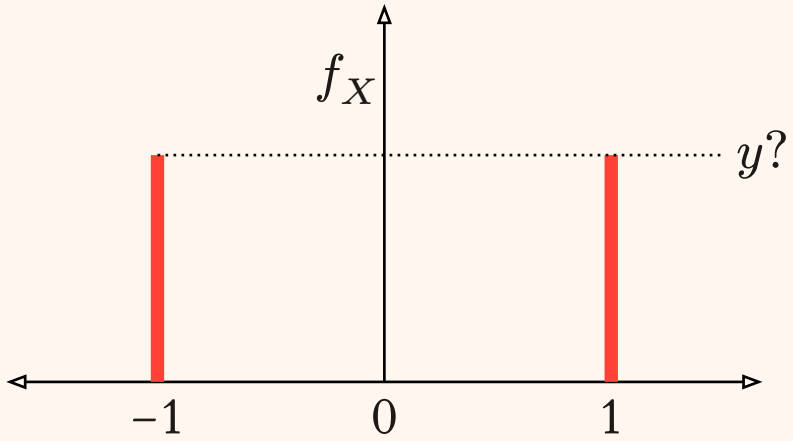
$$\mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$

- calculate for this distribution
- calculate for $Y = 2X$ at home

Population statistic III: variance

Goal: measure *spread* – average distance from mean

A toy distribution



Try 1: $\mathbb{E}[(X - \mathbb{E}[X])?$

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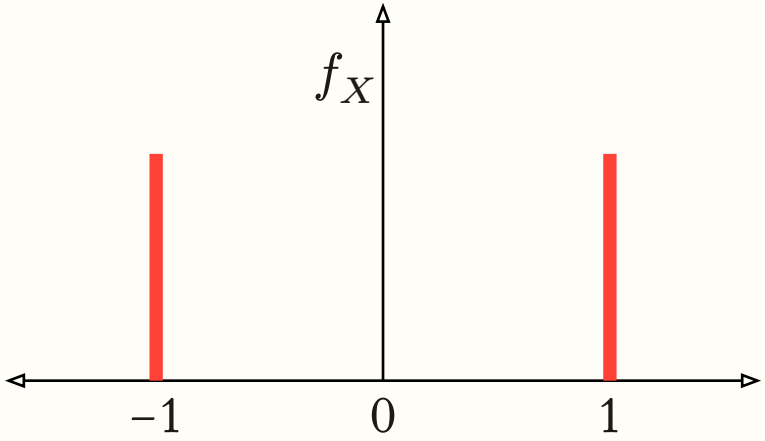
Also variance

$$\mathbb{E}[X^2] - \mathbb{E}[X]^2$$

- Derive for yourself with cheatsheet

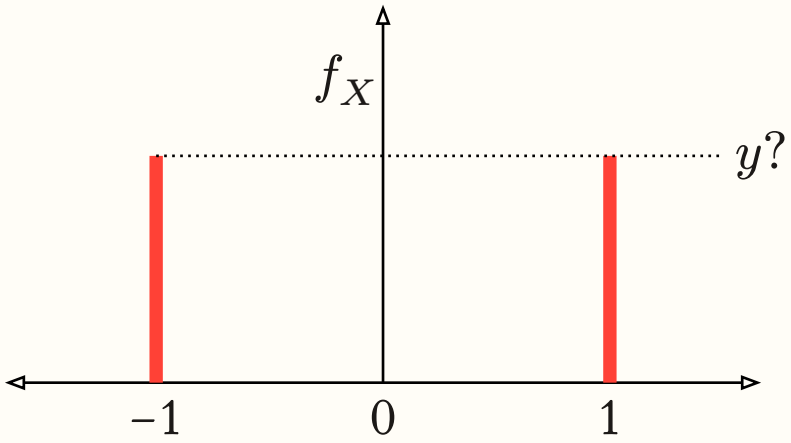
How does variance scale?

PMF of X

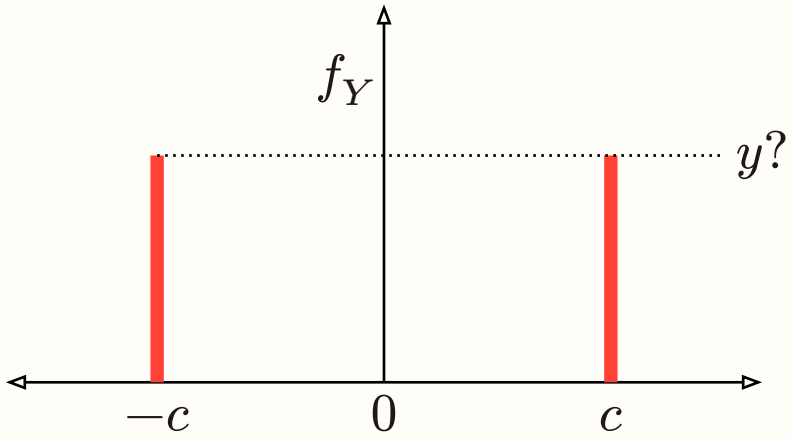


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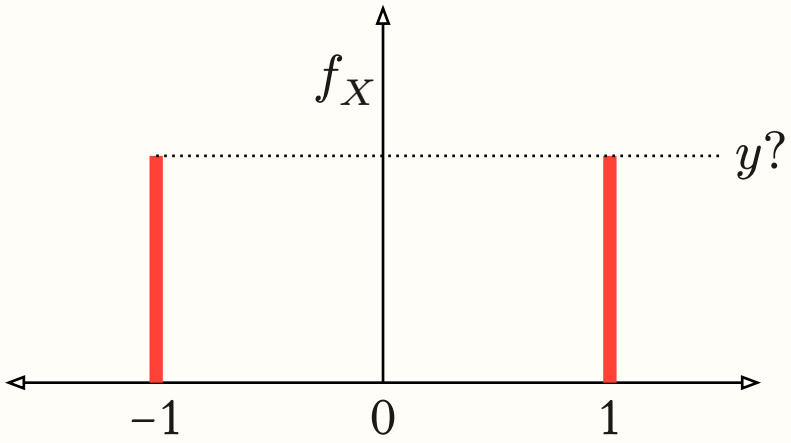
PMF of $Y = cX$ (constant $c \in \mathbb{R}$)



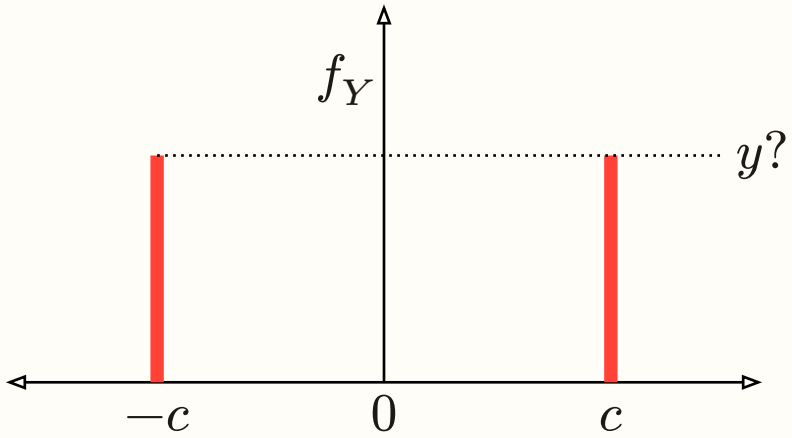
$\mathbb{E}[Y]?$ $\mathbb{E}\left[(Y - \mathbb{E}[Y])^2\right]?$

How does variance scale?

PMF of X



PMF of $Y = cX$ (constant $c \in \mathbb{R}$)



$\mathbb{E}[Y]?$ $\mathbb{E}[(Y - \mathbb{E}[Y])^2]?$

$$\operatorname{var}(cX) = c^2 \operatorname{var}(X)$$

$$\operatorname{var}(X) + \operatorname{var}(Y) = \operatorname{var}(X + Y)$$

(for independent X, Y)

Back to real life and limited data[1]

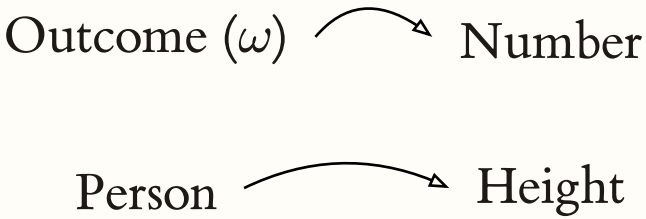
[1] which previous population statistics are not

Back to measuring heights at Sussex

Experiments

- Measure 100 people's heights at Sussex
- Average was 172cm

Random Variable $X(\omega)$ per experiment:



Each height is a sample from a single experiment:

$$X_1, X_2, ..., X_{100}$$

Sample statistics

- Summarise distribution from samples (possible)
- Not population (impossible)

Sample statistics run-through

Data collection

- Run experiment n times
- Sample random variable X

Sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Sample variance

$$s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}_n\right)^2$$

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These are the same summaries as before!

- But computed on (available) data, not (unavailable) distribution

Recap: two worlds

	Population	Sample
Middle	Expectation μ (fixed, unknown)	Sample mean $\bar{X}_n$ (known, random)
Spread	Variance σ^2	Sample variance s^2
Shape	The PDF/PMF	Histogram

1. Population statistic:

- Chosen by universe
- Unknowable

2. Sample statistic:

- Chosen by luck of experiment

Statistics

- Use 2 to say something about 1

Sample statistics are themselves random variables

Experiments

- Measure 100 people's heights at Sussex
- $\bar{X}_{100} = 172\text{cm}$

What if we took data again?

- e.g. $\bar{X}_{100} = 173$ cm

Sample statistics are themselves random variables

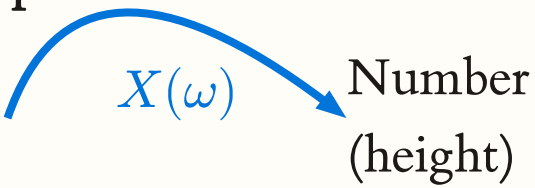
Experiments

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- e.g. $\bar{X}_{100} = 173$ cm

Outcome experiment



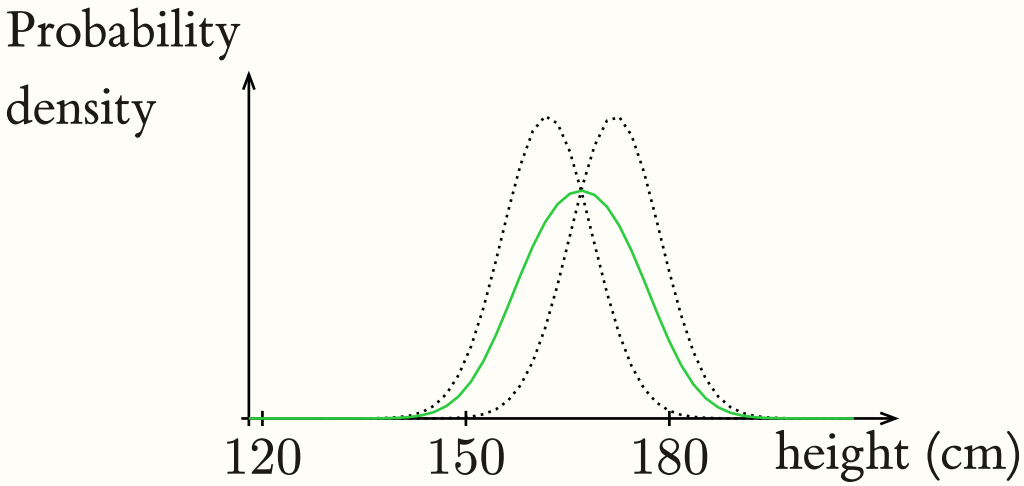
n-sample experiment



What's the distribution of $\bar{X}_n$?

- Useful to characterise its *spread*

Outcome experiment



Distribution of n-sample mean

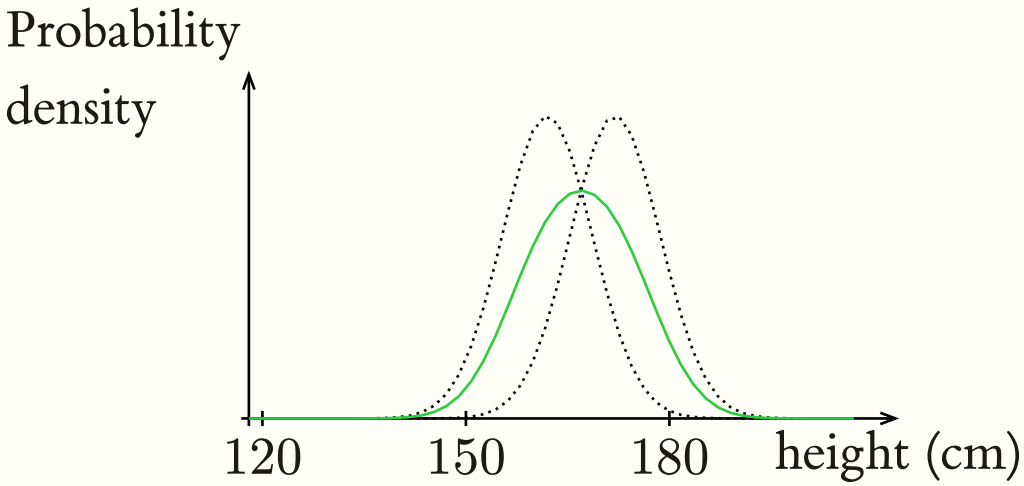
Poll

Distribution of average of $n = 60$ heights:

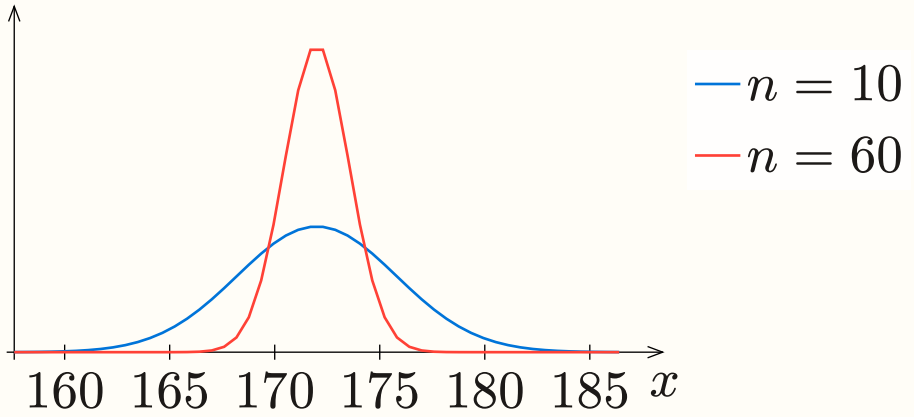
- Bimodal? Gaussian? Other?

Notebook demo

Outcome experiment



Distribution of n-sample mean



Poll

Distribution of average of $n = 60$ heights:

- Bimodal? Gaussian? Other?

This was inevitable

- Regardless of X distribution
- Why…?

Central Limit Theorem

Suppose $\{X_i\}_{i=1}^n$ are independent, identically distributed (i.i.d) variables[1]

Define their true, shared, population mean and variance as μ, σ^2

We don't need to know their distribution

[1] e.g. n repeats of the same experiment

[2] $\xrightarrow[d]{}$ means *converges in distribution*

Central Limit Theorem

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We don't need to know their distribution

Then[2]

$$\bar{X}_n \underset{d}{\to} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

[1] e.g. n repeats of the same experiment

[2] $\underset{d}{\to}$ means *converges in distribution*

Central Limit Theorem

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Then[2]

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Caveat

- n how big for enough convergence?
- Unknowable, depends on X distribution! Culture: $n = 30$

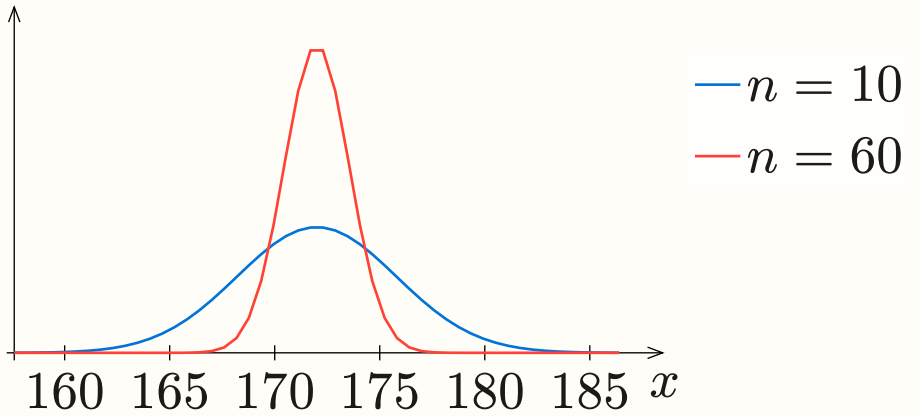
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[2] $\underset{d}{\to}$ means *converges in distribution*

Central Limit Theorem

CLT, again

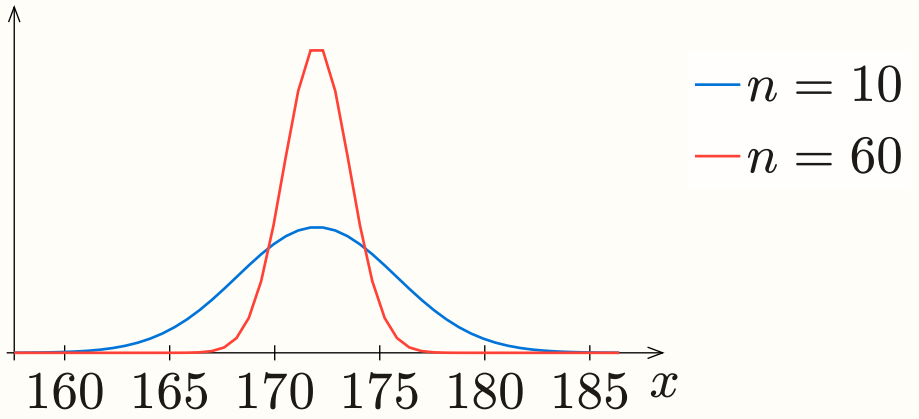
$$\bar{X}_n \xrightarrow[d]{} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$



Central Limit Theorem

CLT, again

$$\bar{X}_n \xrightarrow[d]{} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$



Population statistics (unknowable)

μ: number you want

σ: spread of *individuals*
(unknowable: cheat in week 5)

What you control

n: number of datapoints

$\frac{\sigma}{\sqrt{n}}$: standard deviation of sample mean

Statistics is extrapolating from limited data

Experiment

- Measure 100 people's heights at Sussex
- Average was 172cm

Claim A

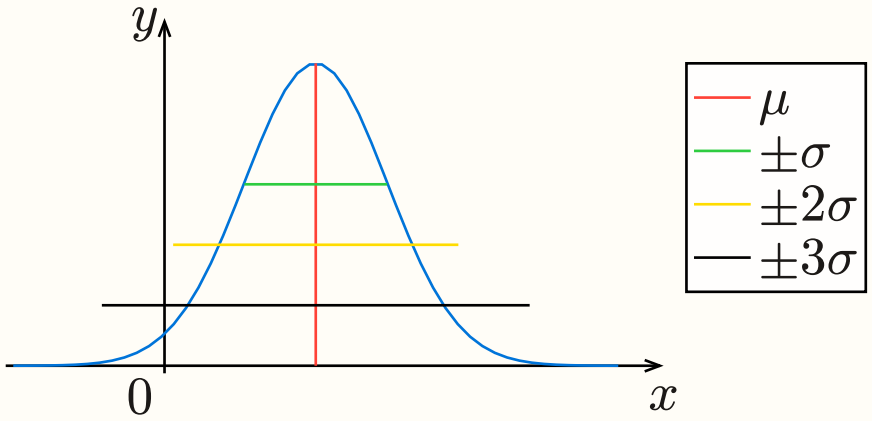
- Average Sussex uni height is inside 172 ± 2.4cm

Claim B

- Average UK height is inside 172 ± 2.4cm

- Claim A plausible?

Guesstimating height σ



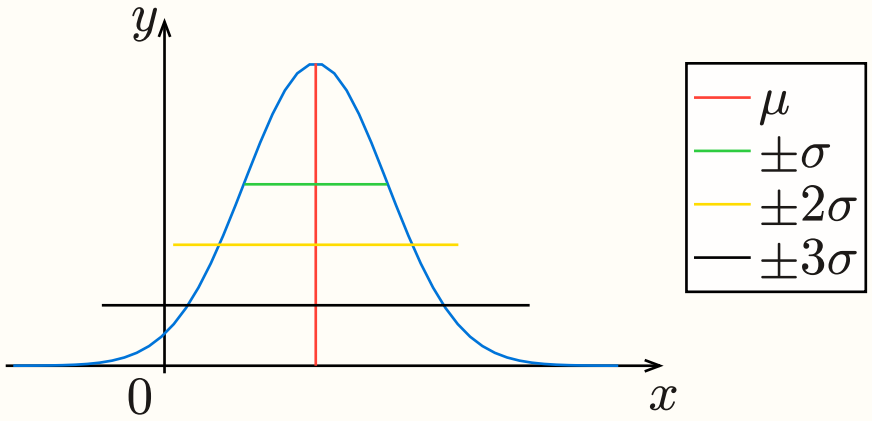
Established height is not Gaussian

Nevertheless good to guesstimate σ

Important table

Number of stds	Probability within
$\pm\sigma$	0.68
$\pm 2\sigma$	0.95
$\pm 3\sigma$	0.997

Guesstimating height σ



Established height is not Gaussian

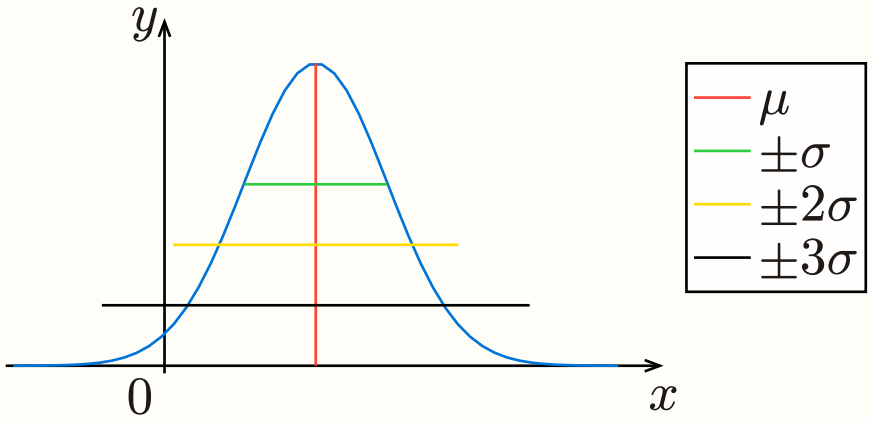
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68% of people are within xxx cm of the mean roughly?

Guesstimating height σ



Established height is not Gaussian

Nevertheless good to guesstimate σ

Important table

Number of stds	Probability within
$\pm\sigma$	0.68
$\pm2\sigma$	0.95
$\pm3\sigma$	0.997

68% of people are within xxx cm of the mean roughly?

eg. $\sigma = 12$cm

Statistics is extrapolating from limited data

Experiment

- Measure 100 people's heights at Sussex
- Average was 172cm

Claim A

- Average Sussex uni height is inside 172 ± 2.4cm

Claim B

- Average UK height is inside 172 ± 2.4cm

Claim A plausible?

CLT

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

- Standard deviation of sample mean is $\frac{\sigma}{\sqrt{n}} = \frac{\sigma}{10} \approx 1.2$cm

Claim A

Probably (95% guesstimate) within 2.4cm of μ!

Statistics is extrapolating from limited data

Experiment

- Measure 100 people's heights at Sussex
- Average was 172cm

Claim A

- Average Sussex uni height is inside 172 ± 2.4cm

Claim B plausible?

CLT

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Are heights at Sussex vs UK i.i.d.?[1]

Reliability of claim is subjective

How would you test?

[1] remember acronym: independent, identically distributed

Oops we measured outside the basketball court

CLT assumptions

Independent:
Next measurement correlated with previous one?

Identically distributed:

- To each other? Fair enough!
- To entire university?

CLT

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

- μ will be sample mean outside basketball court!